Engineering Chemistry · Unit_02 · Electrochemistry and corrosion

DRAFT — UE26CY151A · Unit_02 — Electrochemistry and corrosion · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

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· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_02 · Electrochemistry and corrosion

1 UE26CY151A · Unit_02: Electrochemistry and corrosion

This unit master organizes source material for Engineering Chemistry, Unit_02, officially titled Electrochemistry and corrosion. It includes 14 video transcripts and 12 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26CY151A · Unit_02: Electrochemistry and corrosion

Official catalogue entry: UE26CY151A, Unit_02 (Electrochemistry and corrosion). Bucket folder present: yes. Videos: 14; slide decks: 12.

EV0000001

2 Origin of electrode potential, cell potential

Study focus: Origin of electrode potential, cell potential. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Today we will start with a new module on electrochemical equilibrium.

Key details

- In this module we are going to study about electrode potential and cell potential.
- We will derive the Nernst equation.
- Then we discuss the different types of electrodes.
- We will look at reference electrodes and also talk about concentration cells.

- And finally, we will discuss about ion selective electrodes.
- In today's class we will discuss the origin of electrode potential, we will talk about cell potential and we will derive Nernst equation.
- Electrochemical equilibria deals with electrochemistry, which is a branch of chemistry which deals with the interconversion of chemical energy and electrical energy.
- In the galvanic cell, we have chemical energy getting converted into electrical energy.
- It is the energy in a chemical reaction.
- Now the reaction has to be a spontaneous reaction because we are not going to provide any energy from outside.
- The chemical energy has to produce electrical energy.
- And since we are going to produce electrical energy, that is power, there has to be exchange of electrons.
- Therefore, we have to have detox reaction.
- Examples of galvanic cell would be in batteries and fuel cells.
- We convert electrical energy to chemical energy.
- Electrical energy is provided externally and.
- We want to get chemical energy, meaning we want to make a chemical reaction happen.
- We require a non-spontaneous redox reaction in an electrolytic cell.
- We use such cells in electroplating, electro refining, and many other metallurgical processes.
- Whenever we do electrochemical studies, the first thing we require is a redox reaction because we are talking of conversion of chemical energy into electrical energy or electrical energy into chemical energy, and both require an exchange of electrons.
- And one thing we should always remember that oxidation and reduction have to happen simultaneously.
- The second requirement is that of electrodes.
- These oxidation and reduction reactions have to happen at the anode and cathode respectively.
- And the third requirement would be electrolytic conductance.
- The conductance is through the movement of ions.
- We have to remember that electrons cannot move through the electrolyte.
- It's only the ions which are going to cause electrolytic conductance.
- We can also have molten electrolytes.
- Apart from that.
- There is a third category of electrolytes which are called solid electrolytes.
- That is the criteria for an electrolyte.
- What happens is.

Example or application from the lecture

- Suppose we have sodium chloride, we know that when we melt it, then the ions become free to move, and therefore we can use molten electrolyte also in galvanic cells or electrolytic cells.
- For example, we have zirconium oxide when it is doped with Y2O3.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

Oxide ion vacancies are created and this oxide ion can move through the solid electrolyte, therefore it acts as an electrolyte.

Key details

- Since it's allowing for the movement of the oxide ions, we can say that zirconium oxide dope by two or three is also an electrolyte.
- When a metal rod is dipped in a solution of its own ions.
- At the interface there is a potential which is developed and this is known as electrode potential and this happens at the interface of the metal and the solution and it is denoted by capital E.
- The first one is ionization or oxidation the metal rod can dissolve and get changed into Mn plus thereby giving out electrons.
- The ionization reaction is faster than the deposition reaction.
- Just now I mentioned that whenever we have a metal rod dipped in a solution of its own ions, there are two possibilities.
- Either the metal gets ionized or the metal ions take up electrons and yet deposit.
- That is M is becoming Mn plus the N electrons.
- We have M and we have the solution which contains Mn plus.
- When this reaction takes place we know that the metal is getting ionized.
- But what happens to the electrons?
- They are left back on the surface of the electrode, and therefore the electrode acquires a negative charge.
- This at the surface of the electrode we have negative charge, and because there is a negative charge on the electrode it is going to attract the positive ions from the solution.
- We have a positive layer which is getting formed, and because of this we see that at the interface.
- There is an electric double layer which is getting formed.
- This electric double layer is responsible for the electrode potential and it is known as the hellholes electric double layer.
- metal rod and it is dipped in a solution of Mn plus.
- But this is case two where the deposition reaction is faster, meaning the reduction reaction is happening faster.
- And get deposited.
- But where are the electrons coming from?
- They are coming from the metal rod.
- Again this electrical double layer is responsible for the electrode potential at the interface and this electrical double layer is known as the.
- And molds electrical double layer.

Example or application from the lecture

- Let us now discuss the origin of electrode potential.
- Let us take up a case where.
- Now suppose the first reaction is faster where wherein the ionization reaction is faster.

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EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

or the reduction reaction that is deposition happens to be the faster reaction.

Key details

- In both cases we understand that there's going to be an electrical double layer which is formed at the interface between the metal rod and the solution and this is known as the Helmor's electrical double layer and responsible for the electrode potential which we see in this system.
- Whether the oxidation reaction will happen faster or the reduction reaction will happen faster is going to depend on the nature of the metal.
- If the metal has greater tendency to undergo oxidation, we find that ionization will be the faster reaction and if.
- The ions have a greater tendency to undergo reduction, then we find that the deposition reaction will be happening at a faster rate.
- So that means it clearly depends on the nature of the metal, and it also depends on the concentration of the ions which are being used, it depends on the temperature.
- And if gases are involved on the pressure.
- If we want to compare electrodes, then we need to define a new quantity which is known as the standard electrode potential.
- It is defined as the potential developed at the interface of the metal in the solution.
- When the metal is in contact with the solution of its own ions, but now we have placed a restriction that the concentration should be unity and the temperature is now fixed at two ninety eight Kelvin and in the.
- In case there are gases involved, then the pressure of the gas also has to be maintained at one atmosphere pressure.
- But single electrode potentials cannot be measured.
- What we can measure is the potential difference.
- Therefore, we need to form a cell where two electrodes are coupled together and such a cell would be called as an electrochemical cell.
- And therefore we see.
- We know that in a Daniel cell what is it?
- Zinc is dipped in a solution of zinc sulphate, copper is dipped in a solution of copper sulphate.
- Zinc rod is dipped in a solution of zinc sulfate.
- We always write the anode first.
- Where oxidation happens, the anode is written first.
- Between zinc and copper we find that the zinc electrode will undergo oxidation.

- The value in this parenthesis is the concentration of the zinc ions.
- So anode is written first and this single line over here represents the phase separation.
- So zinc is solid and Zn two plus is in the liquid form, therefore there is a phase separation shown by the single line.
- Again is separated by a single line, but and this is where this is the cathode.
- So we have the anode first and then we write the cathode, and in between we can see these double lines.
- This represents the salt bridge.
- What we have to remember is we cannot write.
- Zinc Zn2 plus and salt bridge copper and then CU2 plus.
- Because the salt bridge is connecting the solutions.
- Therefore, this is the representation and this is known as cell notation.
- The difference in electrode potentials of the electrode is something which we can measure and that is known as the cell potential.
- Potential and it is denoted by e cell.
- Again, cell potential is going to depend on the concentrations of the solutions used and also on the temperature and also on the pressure as well as the electrodes which have been used.
- So therefore, in order to.

Example or application from the lecture

- Suppose we take the example of Daniel Cell.

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EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

compare we need to have a standard value which is known as the standard electrode potential and e cell depends on the concentration temperature and partial pressure of the gases involved.

Key details

- Cell potential is.
- Because we know the cell is made of two electrodes, one electrode is the anode electrode, the other electrode is the cathode electrode.
- And also I mentioned that anode is written first and then that is on the left hand side and cathode is written on the right hand side of the salt bridge.
- Because it represents the driving force for the cell reaction to take place.
- What does it tell us that for any spontaneous reaction we know that delta G has to be negative and therefore E cell should be positive.
- And if the reaction is non-spontaneous, then delta G will be positive and therefore the E cell has to be negative.
- The standard reduction potentials have been listed in a table in ascending order.
- This is to help us predict the electrochemical behavior of an electrode electrolyte system.
- This series is known as the electrochemical series.

- Most often we find that it is the standard reduction potentials which have been used and such an arrangement is referred to as the electrochemical series.
- This series is going to be very informative.
- A negative value of standard reduction potential indicates oxidation tendency while a positive value is going to indicate a reduction tendency.
- Whereas copper we find that.
- CU2 plus taking up electrons giving you copper that is the reduction potential is plus 0.34.
- Hence the value is positive.
- Electrochemical series lists the standard electrode potentials which are measured under standard conditions of unit concentration two ninety eight Kelvin temperature and one atmosphere pressure.
- But most of the times we do not deal with standard conditions.
- Now the reaction at the electrode is the reduction reaction, so we write it as Mn plus N electron is equal to M because we are talking of reduction potentials.
- Written as minus delta G is equal to W max.
- The system we are talking about here is a metal rod dipped in a solution of its own ions.

Example or application from the lecture

- E cell is given by electrode potential of the right hand side electrode minus the electrode potential of the left hand side electrode or in other words we can say e cathode minus the e anode.
- because the change in free energy for any reaction any system tries to go to minimum free energy so the change in free energy is given by minus nf ec.
- The of the two electrodes, and that will be the driving force for the reaction to take place, the difference in the free the electrode potentials will decide the reaction the they decide the reaction and in other words thermodynamically that is expressed as delta g is equal to minus nfe c let us talk about the electrochemical series.

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EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

For any electrical chemical system, the maximum work done is given by the total charge available multiplied by the energy available per unit charge.

Key details

- The total charge available here is equal to the number of moles of electrons exchanged in the redox reaction, that is n we can see here, multiplied by the charge carried per mole of electrons, and that is f.
- Therefore, now the W max becomes equal to N F.
- multiplied by E.
- That is the total charge available is Nf and the energy available per unit charge is E, therefore W max is equal to N F E and therefore delta G becomes equal to minus N FE by substituting in this equation.

- Under standard conditions, delta G is delta G naught that is equal to N minus of Nfe.
- The reaction we are talking about is Mn plus N electron is giving you M.
- What is q?
- It is the reaction quotient.
- We know what is delta G naught and we also know Q.
- In this equation, we are going to substitute, we get minus of Nfe is equal to minus Nfe naught plus R T log M over Mn plus.
- that is 96500 coulombs per mole.
- For pure substances we take M concentration equal to one.
- Substituting for R 8.314 joules per Kelvin per mole, temperature as two ninety-eight Kelvin, F is nine six five zero coulombs per mole.
- multiplied by 2.303 then we get the value 0.0591.
- This is for a single electrode.
- The only thing which is going to change is the reaction taking place.
- Q value is concentration of the.
- Products raised to the respective stechometric coefficients divided by concentration of the reactants raised to the respective stepometric coefficients.
- Delta G remains the same.
- It is minus N F E and Delta G naught is going to be minus N F E naught.

Example or application from the lecture

- Total charge is N into F and the energy available per unit charge is will be electrode potential because electrode potential is given by energy per unit charge.
- That is if you if the unit is volts then that is given by joules per coulomb and that is the electrode potential which is given by E.
- A thermodynamic equation which is going to react which relates the reaction quotient and the decrease in free energy is given by delta g is equal to delta g0 plus r t log q.

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EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 25:00–29:59

cell emf and the standard cell emf values so dividing by minus nf and of course we have to substitute for q the nonst equation at 298 Kelvin now is equal to e naught minus 0.0591 by n log of.

Key details

- C raised to the power of small c d raised to the power of small b divided by a raised to the power of small a and b raised to the power of small b.
- and n is the number of electrons transferred and e0 is the standard EMF of the cell.
- Now in the next class, we are going to discuss about the different types of electrodes.

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EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

3 Types of electrodes

Study focus: Types of electrodes. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000026 · EV0000027

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module four on electrochemical equilibria and in today's class we are going to discuss about the different types of electrodes.

Key details

- The metal ion electrode, the metal insoluble salt ion electrode, the gas electrode, amalgam electrode, redox electrode and iron selective electrode.
- In order to form a cell, we need two half cells, and therefore two electrodes are required.
- The first one is a metal ion electrode.
- We have a metal rod which is in contact with a solution of its own ions.
- Remember that we always write the reduction reaction because we all talk about reduction potentials.
- Since a potential is formed at the.
- Electrode and the solution interface.
- This is a familiar Nant equation which we have already learnt.
- M of MN plus is equal to E naught that is the standard electrode potential of M Mn plus two point three zero three R T over NF log of Mn plus.
- So that is the first type of electrode.
- How would we describe this electrode?
- As the name suggests, we have a metal which is in contact with a sparingly soluble salt of the metal and that is again dipped in a solution of soluble salt of the same anion.
- Mercury, which is a metal, and that is in contact with a sparingly soluble salt of mercury, that is.
- A G2 Cl2 and now this is dipped in a solution of KCl.
- Because it should be the solution of a soluble salt of the same anion.
- And here what do we have in silver chloride electrode?
- We have the silver metal which is in contact with silver chloride and that is in contact with HCl.
- The equation is AG Cl plus electron is giving you AG plus Cl minus.
- What we have to be careful about is that this is not A G plus and even if we write A G plus the equation would not be balanced.
- Plus Cl minus and we can see this electrode over here in the diagram we have a silver wire which is coated with silver chloride that is the sparingly soluble salt of silver and that is dipped in a solution either of KCl or SCl.

- It should contain it should be a solution containing the anion and the anion of the salt is Cl minus.

Example or application from the lecture

- Let us see what would be the expression for the electrode potential, and that is given by the Nant equation.
- Let us come to the second type where we have metal, metal insoluble salt, iron electrode.
- The most common example for this would be the calomel electrode.

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EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

Core explanation · 05:00–09:59

Zero point zero five nine one by one log of Cl minus.

Key details

- All of you are familiar with the hydrogen electrode.
- We know that in a hydrogen electrode you have this a bell-shaped jar which is being which is used here, and it has an inlet where we are going to let in hydrogen.
- It is usually platinum or it could be a platinum wire or a platinum foil.
- So and then this is now this whole assembly is dipped into a solution of H plus ions because those are the ions to which the hydrogen gas is reversible.
- The platinum wire which we are using, what is the purpose of that?
- It provides the electrical contact and facilitates the establishment of equilibrium between the gas and its ions.
- We have platinum in contact with H2.
- We have to.
- Use chlorine gas which is going to be bubbled around the platinum wire and now this has to be dipped in a solution of Cl minus ions.
- Electrical contact provid facilitates the establishment of equilibrium.
- between the gas and its ion.
- We are talking about this equilibrium.
- Now in the Nernst equation we have E is equal to E naught minus 0.0591 by 2 because here we are using two electrons.
- That's why we have to make sure we write the reduction reaction only.
- But hydrogen is a gas, so we have to use the partial pressure of hydrogen here and.
- What is in the reactant side?
- We have H plus ions.
- When we write the log term, it is partial pressure of hydrogen divided by H plus the concentration.
- Raise to the power of 2.
- So this log term goes to 0, and therefore we get E is equal to E naught.
- The fourth of electrode is an amalgam electrode.
- Now as the name suggested suggests, amalgam means any alloy of Merkle.

- metal metaline electrode the only difference is that the metal is now taken in the form of an amalgam.
- So therefore we have say if we have a lead amalgam, then we use lead in the form of an amalgam and it has to be dipped in a solution of its own ions.
- It is just like the.
- metal ion electrode where we have metal ion taking up electron and becoming the metal.
- But here the only difference is we are taking the metal in the form of an amalgam.

Example or application from the lecture

- Let us come to the third type of electrode that is the gas electrode.
- So hydrogen electrode is a good example for the gas electrode.
- So, example would be hydrogen electrode, as I mentioned just now.

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EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 10:00–14:59

Is very active.

Key details

- If we amalgamate the sodium, then we would have sufficiently reduced the activity of sodium.
- And then we can make this electrode.
- So not only that we can actually control the activity of this electrode because we can decide the percentage of mercury which we are going to add into the metal to the form the amalgam.
- 2 is the number of electrons log of.
- Whenever we have pure metal, the activity is taken as one.
- But here since we have amalgamated it.
- Definitely the activity will not be one.
- In fact it will depend on the percentage of lead which is there in this in this electron.
- Then we have redox electrodes, or they are also known as the oxidation reduction electrode.
- We have say tin.
- Which is in the form of two species Sn2 plus and the Sn four plus.
- When we have both these species present in the solution and we place a platinum wire inside, then we know.
- There is a natural tendency of one of the oxidation states to get converted into other, and in that process electrons are getting exchanged.
- formed and that elect that potential is being picked up by these by the inert electrode.
- The potential arises due to the tendency of one form to change into the other.
- And if we have both these species present in the same solution and we place a platinum.
- And wire there, we will get what we call as a redox electrode.
- Other examples would be platinum C3 plus C4 plus.
- or platinum Sn two plus S N four plus as you can see in the picture over here.

- It would be Sn four plus taking up electrons and forming S N two plus the Nernst equation will be based on this equation now.
- ELE is equal to E naught minus 0.0591 by 2 log of S N2 plus that is the products divided by the reactants.
- A special case of redox electrode is the quin hydron electrode.
- It consists of an inert metal such as platinum which is immersed in a solution containing quinone and hydroquinone.
- We can see here this is the reaction which is happening.
- It involves an exchange of electrons in the presence of H plus ion.
- Quinone and hydroquinone present together in the solution.
- And if we place a platinum electrode there, we will be able to get.

Example or application from the lecture

- So a good example of that would be your iron, which exists as Fe two plus and Fe three plus.

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EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 15:00–19:59

How do we represent Q H two and divided by that of quinone that is.

Key details

- Q concentration and we also have H plus.
- These are also referred to as membrane electrodes.
- An ion selective electrode, as the name suggests, it is going to select a particular ion in the presence of other ions.
- and it is going to tell us the concentration of that particular ion or in other term in other words it is going to develop a potential based on the concentration of that one particular ion.
- It is because it has a membrane which is in contact with the solution, and this membrane is the most important constituent, and this membrane is being able to exchange the ions and since it is being able to exchange the ions it can it develops a potential and that.
- That is why it can detect the concentration of one ion in the presence of other ions.
- And what are pH meters which tell us the concentration of electricity.
- H plus so the glass membrane is selective to H plus ion.
- Because it is selective to H plus ion, it can give us the concentration of H plus N, or in other words, it can help us determine the pH of a solution.
- And it is also sensitive to not only H plus, sometimes to NE plus and K plus.
- Pla E no E G would be equal to E naught G plus 0.0591 log of H plus Ion.
- Extremely sensitive and they are also known as membrane electrodes because membrane is the most important constituent of a glass electrode.
- And in the next class we are going to talk about reference electrodes.

Example or application from the lecture

- So, an example for this would be glass electrode, and we know that glass electrode is used in pH meters.

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EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

4 Reference electrodes, calomel electrode

Study focus: Reference electrodes, calomel electrode. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000042 · EV0000043

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We'll continue with module 4 on electrochemical equilibrium.

Key details

- And in today's class we are going to look at reference electrodes.
- That would be the primary reference electrode which is your standard hydrogen electrode and then we will study about the secondary reference electrodes which are galomyl electrode and silver chloride electrode.
- We know from the previous classes that whenever we form a cell, it will be two electrodes which are coupled together.
- What we can measure is just the potential difference.
- and measuring the electrode cell potential we will be able to determine the potential of the other electrode.
- This other electrode is known as the indicator electrode.
- Then we combine the reference electrode with the indicator electrode.
- You're familiar with this, they are called the primary reference electrode and the secondary reference electrode.
- And for secondary reference electrodes we have.
- Calamile electrodes and the elect and the silver chloride electrode.
- As I mentioned just now, the standard hydrogen electrode, which is also referred to as the SHE, is used as a primary reference electrode.
- It is assigned a value of 0.0 volts.
- It is by convention and internationally we accept that electrode potential for a standard hydrogen electrode is assigned a value.
- of 0.0 volts.
- We have a gas being bubbled around an inert electrode which is dipped in a solution of ions to which the gas is reversible.
- as we studied yesterday.

- Always one atmosphere and the H plus concentration is maintained at one molar concentration.
- As mentioned earlier, we have to form a cell.
- Zinc rod dipped in Z2 plus ions.
- How do I know this is a standard hydrogen electrode?
- Because I've used one atmosphere pressure and one molar concentration of H plus ions.
- And we know that E cell is always E right hand side minus the E left hand side, which is E cathode minus E anode.

Example or application from the lecture

- Let us first define reference electrodes.
- An example for primary reference electrode in fact this is the only electrode which is used as a primary reference electrode is the standard hydrogen electrode.
- So let us look at first what is the primary reference electrode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 05:00–09:59

And this is equal to 0, as I said mentioned just now that standard hydrogen electrode potential is always given a value of 0 minus E of the zinc electrode.

Key details

- Since it is a positive value, I know that the connections I have made here are right, meaning what that zinc electrode will act as the anode and the hydrogen electrode is going to act as a cathode.
- If there was any mistake in this, if I had put hydrogen electrode here and the zinc electrode here then I would have measured E cell as a negative value but since I got a positive value I know that zinc will act as anode here and hydrogen electrode will act as the cathode.
- Zn Zn2 plus.
- Are also at one molar concentration.
- Nevertheless, from this arrangement, it can be any concentration of zinc, doesn't matter.
- I can get the value of the electrode potential if I can measure the cell potential.
- Any electrode you want to measure the potential, whatever cell potential you measure is the value which is to be assigned to that particular electrode.
- Why do we need any other type of reference electrode?
- Continuously to maintain it at that particular value is very difficult.
- So that is the first problem we face, and then we have platinum.
- Which is highly susceptible to poisoning.
- We know that platinum gets poisoned easily by the different impurities which could be present in the gas, and the third point is that the high standard hydrogen electrode cannot be used wherever there is an oxidizing and reducing energy.
- Therefore, these are the problems associated or disadvantages or limitations of the standard hydrogen electrode.

- Hence we have to look for other type of electrodes which will act as reference electrode.
- Now for a secondary reference electrode because we have these limitations with the standard hydrogen electrode, all these problems should be removed now.
- Apart from that.
- Of course, this is very important, the electrode potential should be accurately known and it should remain stable.
- for a long period of time.
- Over a long period of time we should have a stable potential so that when we measure the potential difference we know that this electrode has this potential and therefore the other part the difference of this.

Example or application from the lecture

- The reason is that the standard hydrogen electrode has certain disadvantages so what are the disadvantages let us have a look at it now maintaining H plus ions at one molar concentration and the pressure of hydrogen at one atmosphere is very difficult so.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 10:00–14:59

potential difference from the electrode potential of the reference electrode would give me the electrode potential of the indicator electrode so one thing it should be accurately known second it should be stable over a long period of time and it can be easily assembled so these are the three things which we look for when we design or select a secondary reference electrode.

Key details

- And with this, of course, since it is a reference electrode, my definition gets complete when I say electrode potential of other electrodes can be assigned with the help of this electrode.
- Come under this category one is the caramel electrode, the other is the silver chloride electrode.
- Now the first thing if you are asked to describe the Calomel electrode and explain the Calomar electrode first thing we need to write or we need to say is that it is used as a secondary reference electrode and it is a very widely used secondary reference electrode.
- And then the next thing, what category does this electrode fall in?
- Metal, insoluble salt.
- Because in a calomyl electrode I know.
- I have to use a metal that is mercury is used, then I need to use H2Cl2, which is the insoluble salt or sparingly soluble salt of the metal, and then I need to use the KCl solution.
- And we place a layer of mercury in it.
- Above this layer.
- We take a paste of mercury and HG2Cl2.
- Hg and over that now we place KCl solution.
- You can see a platinum wire is dipped here and it will go into the mercury.
- Platinum wire, it is providing the electrical contact.

- Mercury being a liquid, we don't have an electric contact electrical contact established here.
- outlets or inlets over here.
- This inlet is for introducing KCl solution as in when we need to replace it and this is connecting to the salt bridge.
- This tube over here outlet for the solvent this is connecting to the solder so this is the construction or this is what your calomel electrode looks like.
- Hg, H2, Cl2 solid that is and Cl minus.
- Because that is true for any electrode, it always depends on the other electrode which is being combined.
- It reacts with the chloride ions which we have provided there in the form of KCl and this gives you HE2Cl2.
- What is it?
- We add this to and what we get 2 H G plus 2 Cl minus is equal 2 Cl minus is equal to H E two Cl2 plus 2 electrons.

Example or application from the lecture

- So we see that mercury first forms a G22 giving out electron because it's an anode so electrons are being given out over here.
- This is the reaction when calomel acts as anode, and you can see electrons are being given out here.

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EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 15:00–19:59

is giving you 2 Hg and then H2Cl2 undergoes dissociation to compensate for the G2 plus which have been used up here.

Key details

- When the caromel electrode acts as cathode, again it is true we can understand because two electrons here, the electrons are being taken up, so it's a reduction reaction.
- Writing the Nernst equation.
- For the Nernst equation, what do we need?
- We need the Q value first.
- We can see H G raised to the power of 2, Cl minus raised to the power of 2 divided by HE2 CL2.
- Are pure form, so the value will be one for them.
- Therefore, the q value is just Cl minus squared.
- Okay, so one now we write the Nest equation E is equal to E naught minus two point three zero three R T over two f two is because of these two electrons over here log of Cn minus square.
- We get the expression that is E is equal to E naught minus 0.0591 log of Cl minus.
- When we get this expression, what we say is the electrode is reversible to chloride ions.
- What does it mean?

- That the electrode potential.
- Is depending on the chloride ions, or in other words, if we know the electrode potential of this electrode, we will be able to determine the chloride ion concentration.
- It's very obvious from this equation E is equal to E naught minus 0.059 1 log of Cl minus so electrode potential depends on the chloride ions.
- But we did not specify what is the concentration of KCl we are using.
- So since now we understand that the electrode potential is going to depend on the chloride ion concentration, that means the KCl which is there in the electrode.
- The potential is going to depend on the concentration of the KCL which we have used.
- If we use 0.1 molar KCl or 0.1 normal, then it will be known as a decinormal electrode and the potential is 0.3358 volts.
- And if we use a one molar concentration on one normal, then it is a called a normal electrode and it is 0.2824 volts, and whereas if it is a saturated solution of KCl and then it is known as a saturated calom electrode and the electrode potential is 2244 volts.
- Silver chloride electrode again is widely used as a reference electrode.
- So that's the first thing we need to note.
- Because mercury being a liquid, we took a tube and we placed mercury.
- And then in that we place a platinum wire for electrical contact.
- But here we don't need to do that because silver is already in the solid form.
- Or we can have a silver clore coated platinum wire.
- It is a metal insoluble salt ion electrode.

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EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

Core explanation · 20:00–24:59

AGCl that is silver chloride which is electrolytically coated over the silver wire and now we need KCl solution.

Key details

- H Cl is also allowed, we can use HCl also of known concentration.
- So Ag plus electron is giving you AG and AG Cl gives you Ag plus Cl minus and therefore overall reaction would be A G Cl plus electron is giving you AG plus Cl minus.
- If you remember when I taught second type of electrodes that is metal insoluble salt.
- Ion electrode I gave this reaction over there and also mention it is not AG plus it is AG and it is balanced with the negative charge is balanced in this way so now we know how we got this reaction so it is ag Cl plus electron is giving you.
- A G plus the Cl minus.
- N is 1.
- potential is going to depend on the chloride ion concentration.
- Based on that we have three different types of silver chloride electrode because again why the chloride ion concentration is coming from KCl or if you have used HCl then HCl so based on the chloride ion concentration if it is 0.1 normal Kl.

- A Cl which you have used, then it is known as a decnormal electrode and the value is electrode potential at 298 Kelvin is 0.289 volts.
- If you use one normal, then it is known as a normal electrode and the electrode potential is 0.223 volts.
- And if you use a saturated solution of Caseyl then it will it will be referred to as it will be called saturated silver chloride electrode.
- And the potential is going to be 0.199 volts.
- Silver chloride electrode also depends on the chloride ion concentration which we have used.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

5 Concentration cells, types of concentration cells

Study focus: Concentration cells, types of concentration cells. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000062 · EV0000063

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We'll continue with module 4 on electrochemical equilibrium.

Key details

- And in today's class we are going to talk about concentration cells, the different types of concentration cells, that is, electrolyte concentration cells and electrode concentration cells and then we study about ion selective electrodes.
- The different types of ion selective electrodes, and then we derive an expression for the electrode potential of an ion selective electrode.
- An electrochemical cell in which identical electrodes are in contact with the solution of identical species but of different concentrations.
- What does this mean?
- We know that the potential developed for this is going to be different from the one where the solution was of a dilute concentration.
- That we know from Nernst equation.
- electrodes and we have a salt bridge for electrical continuity, we find that the potential difference can be measured and this whole setup becomes of a galvanic cell.
- But each of these electrode is made of the same electrody material.
- The solutions are also of the same electrolyte, but the only difference is in the concentration of the.
- We have taken a cell which has two copper electrodes and they are immersed in copper sulphate solutions such that C2 is greater than C1.
- One of the concentrations is more than the other.

- And as usual, like any galvanic cell, oxidation will take place at the anode and reduction will take place at the cathode.
- The cell will be represented like this Cu, C U2 plus there's a soil bridge, and then we have again C U2 plus C U.
- The only difference is in the concentration.
- And similarly, at the cathode, we have C U2 plus taking up electrons and forming CO because there is a there has to be a reduction reaction again copper concentration is C2 at cathode is C2.
- Look at this reaction, we have to know what is Q.
- What type of electrode is this?
- I told in the previous classes that.
- This electrode is metal rod dipped in a solution of its own ions.

Example or application from the lecture

- Let us look at this picture.
- Let us take an example just now you saw in the picture.
- So let us say what are we assuming here that the concentration at the anode is C1 because remember cell representation, we always write the anode first and then the sol bridge and then the cathode.

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EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 05:00–09:59

For the anode would also be E naught plus 2.303 RTORNF log of C1.

Key details

- And E cell.
- When you write this expression we get E cell is equal to 2.303 R T over N F log of C2 by C1.
- What is N?
- The number of electrons getting exchanged and this is at.
- Room temperature that is 298 Kelvin.
- What we have to note here is that now E0 from both got cancel the anode and cathode e naught is not there.
- So therefore the expression does not contain e naught.
- The cell potential is going to be positive only when C2 is greater than C1, because.
- The expression was 0.0591 over n log of C2 by C1.
- We know that we took.
- C2 as the cathode concentration and C1 as the anode concentration.
- You can see here, here the E cell will be positive.
- Positive provided the cathode concentration is more than the anode concentration.
- What does it mean?
- If the difference in the.

- Concentrations is very high, then the power which is generated, the EMF, is going to be high.
- If the concentration is not very high, we do not get a very high value of EMF because of the.
- Again, because of the expression where the ratio C2 by C1 is appearing.
- Therefore, if C2 if the ratio is high, then only we get the a good power.
- We can we have a high EMF being generated from this set.
- That the cell potential will be 0.
- When C1 is different from C2, and when will it be positive if C2 is greater than C1.
- This third interpretation, the third point which we discussed just now, when C2 is equal to C1, EMF of the cell becomes equal to 0.
- Finally, the cell wants to come to a condition where EMF of the cell becomes E is zero because that also means.
- Delta G is equal to 0, and what does that mean?
- That the cell has reached equilibrium.
- So what would be the equilibrium condition for a concentration cell?
- C2 becomes equal to C1.
- And since we are saying C2 is a higher concentration, it's very obvious that higher concentration will try to lower the concentration, and that is possible only if a reduction reaction takes place.
- The other case where at the anode we said the C1 concentration is low concentration.
- How will it increase only if the metal dissolves more and comes into the solution.
- Therefore, at the cathode, the concentration should be.
- high as compared to the anode and once these concentrations are equalized then the cell PMF becomes zero.
- So in other words the cell has reach an equilibrium condition.
- As long as there is a difference in the concentration the cell is going to function.

Example or application from the lecture

- is always E cathode minus E anode and all you have to remember here is that we always consider the reduction potential.

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EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 10:00–14:59

Just now what we saw was an electrolyte concentration cell.

Key details

- There's another type of concentration cell which is known as.
- The electrode concentration cell.
- We have the electrolyte concentration cell where we have two same electrodes that are dipped in the same electrolyte but with different concentrations of the electrolyte.
- Would again, since it is a concentration cell, the electrodes are same, the electrolyte is same, but here.

- The activity of the electrode itself is different.
- The concentration of the electrolyte remains the same, but the activity of the electrode itself is different.
- I taught you about sodium amalgam in types under types of electrode.
- This is another type of electrode where the metal is taken in the form of amalgam to reduce the activity of the metal.
- These two electrodes now what is different?
- The electrolyte concentration, if you see there's only one electrolyte.
- The activity of the electrode itself is different.
- at the anode divided by the concentration of the sodium amalgam at anode divided by the concentration of the sodium amalgam at the cathode.
- How do we get to that expression?
- Two sodium amalgam electrodes which are coupled together and both are dipped in a solution of Na plus ions.
- What would be the reaction at anode and cathode?
- We have C2 activity as you can see from the cell representation.
- That's an oxidation reaction.
- At the cathode we have NA plus electron is giving you sodium amalgam.
- Of C2 concentration.
- We are writing the cathode.
- H G C2 divided by NA plus.
- And similarly, the Nernst equation for the anode for the reduction potential will be E naught minus 2.303 R T over N F log of sodium amalgam with C1 concentration by NA plus.
- E right minus e left, that is e cathode minus e anode, and finally you get this expression.
- Then now you see 2.303 R P R N F log of NAHG C1.
- It's the anode concentration and C2 is the cathode concentration.
- It is not that always we have the cathode concentration will should be higher than the anode concentration.
- at the one at the cat hole.
- That for every system we need to derive and see.

Example or application from the lecture

- Just now we saw this example where we have two copper electrodes but of different concentrations of copper sulphate and when they are coupled together we see that EMF is generated and the cell potential is given by 2.303 R T over N F log of C2 by C1.
- Let us concentrate on that.
- Where what is an example for such an electrode?

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EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

Core explanation · 15:00–19:59

What is the final expression?

Key details

- Only then can we suggest whether the anode concentration should be higher or cathode concentration should be higher.
- It is what do we understand?
- It is based on the reactions happening at the anode and cathode.
- Such an electrode concentration cell would be if you have two hydrogen electrodes, you can see here.
- Electrolyte concentration cell, it is an electrode concentration cell because the electrode here, the hydrogen itself has a different partial pressure as compared to the hydrogen at the.
- They are at different partial pressures.
- Electrode concentration cell.
- Again, here we see it is the and the cell potential will be positive provided the.
- What is P1?
- The partial pressure of hydrogen at the anode is higher than that of the cathode.
- That the anode concentration should be high or the cathod concentration should be high.
- If we take a chlorine electrode, like two chlorine electrodes, now obviously the electrolyte is common, so there it is again an electrode concentration cell.
- You are in a log of P2 by P1.
- It is different from what happened for the hydrogen electrode.
- We have to derive and find out what the expression.
- Cathode partial pressure should be higher of chlorine as compared to the partial pressure at the anode.
- We need to determine what is the E cell, what is the expression for E cell, in order to understand when the cell reaction is going to be spontaneous or when E cell will be.
- positive and based on that we can decide the cathode and the anode concentration provided we want a positive emf then we have to see the anode and the con cathode concentration we will decide based on the expression which we get.
- Very selective to a specific ion, and that is why they are so important.
- What does it mean?
- That in the presence of other ions, it can selectively detect one particular ion, not only selectively detect, it can actually tell us.
- The concentration that means it can qualitatively detect the ion ignoring the other ions and it will also tell quantitatively tell us the concentration of that particular ion.
- Ion selected electrodes that is how it is named, and therefore the potential will be since it's an electrode, there will be a potential form, and that is going to be a function of the concentration of that particular ion.
- It is determining concentration of one iron in the presence of other ions, so it can actually act as a chemical sensor.
- They have a membrane.
- How do they achieve that?
- That finding only one ion when there are so many other ions present, it is because of the membrane which is capable of exchanging only that specific ion.
- The most important part is role is played by the membrane.
- It is capable of exchanging the specific ion with the solution with which it is in contact.

- Being able to recognize and detect and also quantitatively tell us the concentration of that particular ion, it is because of the membrane.

Example or application from the lecture

- So another example for.
- So this is again another example for.
- The Nernst equation we can derive and find out that it will be given by 2.303 R T over N F log of P1 by P2.

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EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 20:00–24:59

Glass electrode is an example for a membrane electrode.

Key details

- They are based on the membrane which is being used.
- As I said, membrane is the most important part of the ion selective electrode.
- Being used, there can be different types of ion selective electrodes.
- So the first type would be where we have crystalline.
- We find that this.
- This is selective to fluoride ion.
- It is lanthanum fluoride.
- in the presence of other ions and where would this be used in water pollution there is a lot of fluoride contamination and that is very dangerous.
- Such an ion selective electrode will be used to determine the fluoride ion concentration in the presence of other ions.
- This was single crystal, this is polycrystalline, where you have AG2S which is being used as a membrane, and that is selective to sulfide ions.
- We know that glass electrode, what does it have?
- It has a glass membrane.
- And this glass membrane happens to be selective to H plus ions mostly, but it can also be selective to sodium ions also.
- So glass membrane is the most important constituent of the glass electrode, and since selective to H plus ions, that means.
- We can determine the pH of a solution using a glass electrode.
- And then we can also have immobilized liquid in a rigid polymer.
- that iron exchangers as the name suggests are capable of exchanging ions to this would be very useful in ion selective electrodes so we have electrodes which are capable of detecting calcium and nitrate ions.
- Why is it a function of the concentration of the ion?
- The electrode potential of an ion selective electrode.
- Ion selective electrode.
- where the concentration is C1.

- Concentration of the Mn plus ions which I want to determine using this ion selective electrode.
- Come to the other side, we have the we have a reference solution.
- Determine the concentration.
- It will be the same solution on this side, but this concentration is known to us.
- Since we are calling it a reference solution, internal reference solution, we understand that this concentration is known to us.

Example or application from the lecture

- The best example for this is single crystal electrodes where we have which are selective to fluoride ions.
- For example, glass.
- Let us understand the principle behind how are how is the electrode potential getting formed in an ion selective electrode.

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EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 25:00–29:59

Is what I want to determine.

Key details

- And apart from this, there is also an inner reference electrode.
- We know it is because of the difference in the concentration across the membrane.
- Therefore, the juntial expression will be 2.303 RT over NF.
- Log of C1 by C two.
- Log of C1 and C2 has gone into the C2 has now become the constant.
- K will be.
- Minus, sorry, minus is missed over here, minus 2.303 R T over NF log of C2.
- But if you look at the membrane itself, the total membrane potential is going to be the sum of the junction potential plus the e reference electrode, because we have placed an reference electrode also which is a part of the ion selective electrode.
- What is the role of this reference electrode, the internal reference electrode?
- Moreover, the junction potential.
- It is there, but there is no external contact, no electrical contact by which we can measure this potential.
- Our aim is to find out what is this junction potential.
- Because why we know that junction potential will tell me what is the concentration C1.
- The moment I say reference electrode, I know what is the potential of the internal reference electrode.
- E reference that is the internal reference electrode potential.
- So therefore the role of this reference potential is just to provide me an electrical contact so that I can.
- Find out what is this junction potential.

- substituted here and plus the e reference.
- Pertaining to that particular electrode which is being used.
- And what is E0 M?
- K plus E reference.
- E m is equal to E naught M plus 2.303 R E over N F log of C1, where E0 M is equal to K plus E reference.
- We still cannot determine C1.
- Because this is a single electrode, and that single electrode potentials cannot be measured.
- We are going to club this electrode with another reference electrode.

Example or application from the lecture

- The junction potential is given by 2.303 R T over NF.

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EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 30:00–34:59

That will be known as an external reference electrode.

Key details

- Because we have already used a reference electrode which is part of the in ion selective electrode.
- To make a cell using the membrane electrode, I need another reference electrode, and that will be referred to as the external reference electrode.
- Here it is the membrane electrode and then here on the anode we have the external reference electrode.
- And this is constant or known to us.
- So therefore, from here I can determine what is E membrane.
- E membrane is the membrane potential, and once I know membrane potential, I can always find out what is.
- So therefore, when we are measuring using any potential meter, so basically it is all automated whenever you want to measure the.
- We can know what is the concentration, but what is happening inside is actually the cell potential is being measured and it is being subtracted, it is being.
- subtract the membrane electrode is calculated using this expression.
- And we can get the concentration of the species directly when the electrode is dipped into the solution.
- The membrane which is being used offers a very high resistance.
- Membrane offers a very high resistance and therefore we will we cannot use ordinary potential meters, therefore, special type of potential meters have to be used whenever.
- We want to measure the cell potential where one of the electrode is an ion selective electrode.
- So, because the membrane which we are using offers very high resistance.
- These ion selective electrode they are used to determine concentration of number of cations and anions, such as we have hydrogen ion, we have lithium ion, sodium ion, potassium, etc.,

lead and arsenic also, and then we can also determine anions.
- like cyanide, nitrate, fluorides or sulfide.
- So, in the next class, we are going to study a specific ion selective electrode that is the glass electrode, which is the most profusely used ion selective electrode.

Example or application from the lecture

- Concentration directly concentration is given to us.

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EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

6 Electrochemical theory of corrosion

Study focus: Electrochemical theory of corrosion. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will start with a new module today on corrosion chemistry.

Key details

- Today's class we will understand what is corrosion, the different types of corrosion, the electrochemical theory of corrosion, and finally we will look at corrosion of iron.
- Whenever the metal surface is exposed to the environment, the environment is going to cause this destruction of the metal.
- It could be water or it could be simply moisture in the air, acids, bases, salts, oils, aggressive metal polishes and other types of solid and liquid chemicals.
- And what is important for us to know is that it is a spontaneous process.
- means whenever a metal is exposed to the environment, the reaction is going to be spontaneous.
- We need not supply any energy for the process to take place.
- And that is the reason why the moment the metal is exposed to any of these chemicals it is undergoing corrosion.
- And most importantly, corrosion is a nuisance for us because it is not just the direct metal loss, it is the indirect losses which happen when corrosion takes place.
- There could be oil pipelines which would get burst.
- So all these are going to result in colossal losses, and therefore corrosion becomes very, very important study for us.
- What exactly happens during corrosion?
- We know that pure metal.
- Is getting converted into undesirable products.

- We made structures made of pure metal, but they are getting converted into undesirable products.
- Rusting of iron is something all of us have experienced.
- There is a reddish brown scale which is getting formed on the iron and steel objects.
- And what is this?
- This is actually hydrated ferric oxide.
- And many times in the when you have these copper articles, we see that they get a greenish.
- color on the surface.
- It is due to basic pupric carbonate.
- Another phenomenon which we obser observe and that is due to corrosion.
- It is getting converted into CUCO3 plus C U O H twice.
- which is actually covered with a thin copper skin and it was originally brown in colour as you can see in the picture.
- But with time due to exposure to the environment, this has resulted in the formation of the basic copper carbon.

Example or application from the lecture

- Firstly, let us define what is corrosion.
- For example, it could result in a bridge collapsing, or it could result in air crashes.
- Let us understand.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

Which we saw here, which is known as blue green patina, and as we see it today, it is this bluish green in color.

Key details

- Apart from this, we can also have silver getting tarnished when it is exposed to an H2S atmosphere.
- We know that silver particles they get this black color.
- that the metal is getting converted into a compound which is n not desirable for us.
- Why do metals undergo corrosion?
- Or rather, why is corrosion a spontaneous process?
- What is happening is in nature, most of the metals are in the form of their compounds.
- that is the mineral ores which are available to us, they are in the form of compounds like they could be oxides, sulphides, carbonates, chlorides, and many other compounds.
- Then when we do an extraction process.
- Extraction is always an endothermic process because we are either supplying heat or we are supplying some electrical energy.
- And because any system tries to go back to the lower energy state.
- That's why in nature we find that this process metal getting converted into the compound or any d undesirable product is.

- B is a spontaneous process because when you extracted the metal it you got it to a higher energy state by supplying energy, therefore there's a natural tendency for it to go back to the combined form.
- One is dry corrosion.
- The other is wet corrosion.
- In dry corrosion, the metal is being attacked directly by the gases.
- These gases are attacking metal surface.
- Throughout this chapter we have to.
- Understand that it only affects the surface of the metal, the bulk of the metal does not get affected by corrosion.
- Because the metal has an affinity for these gases.
- Say it could be oxygen.
- chlorine, H, ammonia, any of the gases SO2 which are present in the environment around, they are directly attacking the metal surface and therefore it is giving rise to some a product.
- on the metal surface.
- So, therefore, we say it's a direct chemical attack, and it is called dry corrosion.
- And when this happens, we say that it is a electrochemical attack on the metal.
- Anode and cathode formation at as we studied in galvanic cells.

Example or application from the lecture

- So let us answer the first question.
- Let us understand this.
- This metal has you have given this much energy, it's an endothermic process, so they end up at a higher energy state as compared to the ore from which they were formed.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 10:00–14:59

and cathode on the surface of the metal.

Key details

- We have to remember always that it is happening only on the surface.
- They are of course made of the anode and the cathode.
- What do we have?
- We have this metal.
- So basically, metal is getting converted into a ion that means into its compounds.
- That's how we define corrosion.
- That metal gets converted into an undesired compound.
- We see that the metal is getting corroded and it is going to get destroyed.
- Of course, there are n number of electrons which are getting generated in this process.
- Remember, it is the same metal, and one part of it is acting as the cathode.
- Now at the cathode we have to remember that metal can never undergo reduction because at the cathode what is the process which is supposed to take place?

- It is reduction process.
- The metal can never undergo reduction.
- Because it's already in the lowest reduced form.
- Therefore, it is the environment.
- It will never happen.
- So it is the environment because we know that again I would and cathode oxidation and reduction.
- Should happen simultaneously.
- We cannot have just oxidation happening and no reduction.
- Some species in the environment is going to undergo reduction and that will result in the formation of AN minus.
- Fact we have to remember that the anode part of the metal is getting corroded whereas the cathode part is completely unaffected by corrosion because the metal remains intact.
- It's only the environment which is taking part in the reduction process.
- These electrons are going to move towards the cathode and what would be the electronic conducting media?
- That would be the metal itself.
- And we have the AN minus which got formed at the cathode.
- Between there will be formation of corrosion product.
- These two are going to combine the Mn plus ions and the AN minus ions are going to move towards.
- each other and somewhere in between we will find the formation of the corrosion product.
- And remember why do we need a conducting medium for such a phenomena to happen?
- On the metal surface, whereas these ions need a medium to move, and therefore a conducting medium is what is required for corrosion to happen.
- And we should always remember that as long as the anode and the cathode reaction happens simultaneously.
- Corrosion process is going to continue.
- At the anode, we don't have any other reaction which can take place, but that the metal is getting.
- oxidized and therefore giving out n electrons and we have Mn plus ions which are getting formed.
- But at the cathode, since it is not the metal which is involved in the reaction, you could you are having the environment which is taking part in the reaction.

Example or application from the lecture

- Let us check out what are the reactions which are actually happening.

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EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 15:00–19:59

To simplify, we can understand there can be two types of reaction, either liberation of hydrogen or the absorption of oxygen.

Key details

- What happens when there is liberation of hydrogen?
- And again the medium could be acidic or it could be neutral or alkaline.
- If it is acidic medium, we observe that obviously there will be lot of H plus ions and it's a cathodic reaction so electron will be taken up and we get liberation of hydrogen.
- In neutral or alkaline medium again reduction because we are at the cathode.
- And it is either a neutral or alkaline medium, then this is the reaction.
- If it's an acidic medium this will be the predominant reaction.
- Again we could have in the acidic medium or in the neutral or alkaline medium.
- Absorption of oxygen will obviously happen when there is lot of oxygen present in the atmosphere.
- If it is a neutral or alkyne medium, then again we have water, but now oxygen is getting.
- absorbed with four electrons taking part in the reaction and H minus.
- Add the anode we know that Fe2 plus ions are going to get formed and add the cathode.
- Most common reaction would be in a neutral medium oxygen is getting absorbed.
- Fe becoming Fe two plus two electron and most common would be of course if it's an acidic medium as in this case then we will find that water will be liberated, oxygen is being used and water is liberated.
- But most commonly we find neutral mediums.
- medium and we get 2 H2O plus O2 plus 4 electron is giving you 4 O H minus.
- Now what happens next?
- The Fe2 plus ions which are got which were formed here, and this is the conducting medium as can be seen here.
- This Fe2 plus ions and the OH minus ions move towards one another and we get the formation of FeOH twice.
- somewhere between the anode and the cathode regions.
- You can see here that the corrosion product is getting formed, of course it will be on the surface because corrosion is a surface phenomenon.
- Fe2 plus and OH minus are diffusing towards one another and we get FeOH twice, but we know that rust is not FeOH twice.
- So what happens next?
- Fe2 plus is going to get converted or oxidized into Fe three plus.
- We know that Fe three plus is a more stable state as compared to Fe two plus.
- Two plus has got converted to F three plus and that is yellow dust.
- That is because you provided an oxidizing atmosphere.
- And this is known as black rust.

Example or application from the lecture

- Let us see.
- Let us now look at a specific case and that is corrosion of iron.

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EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 20:00–24:59

of oxygen FE3O4 if is actually Fe O.

Key details

- Fe two O three.
- It is known as ferrosopheric oxide, Fe three O4, and therefore in limited oxygen we will see the formation of black rust.
- So now in the next class we are going to talk about the different types of corrosion.

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EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

7 Numerical problems based on electrode and cell potential determination

Study focus: Numerical problems based on electrode and cell potential determination. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000110 · EV0000111

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We'll continue with module four on electrochemical equilibria and in today's class we are going to discuss some numericals on electrochemistry.

Key details

- First we'll look at some problems on Nernst equation and then we look at a problem on ion selective electrode that is the glass electrode.
- And the values in the parenthesis stand for the concentrations of Fe2 plus ions and AG plus ions.
- The question says write the overall cell reaction and determine E naught cell and E cell at 25 degrees centigrade.
- Standard reduction potential of the Fe two plus Fe electrode that is minus 0.44 volts and the standard reduction potential of AG plus AG that is 0.80 volts.
- First and foremost we need to write the overall cell reaction, iron being at the anode.
- We have to write the reaction as Fe giving you Fe2 plus 2 electron and this being silver being at the cathode, the reaction will be reduction reaction.
- We have multiplied this reaction with 2.
- In order to balance the number of electrons, overall reaction is Fe plus 2A G plus is giving you Fe2 plus 2 ag.
- We know the Nernst equation.

- So from the Nernst equation we will be able to determine E cell because we know what is E naught cell.
- But first and foremost, whenever you have a reaction, when the overall reaction, it's always a good practice to find out what is Q.
- Before you go ahead and substitute in the Narnst equation as done here.
- We have Fe2 plus and AG the whole square that is concentration of the reactants by Fe and AG plus square.
- Now we have not taken AG and FE into consideration because we know that for pure substances.
- The concentration is taken as 1.
- And the answer we get is 1.2193 volts.
- This potential, cell potential at 25 degrees for a concentration cell.
- One is the electrode type, the other one is the electrolyte type.
- We see that the electrolyte concentration is just one, but the activity of the electrode itself is different.
- We are calling it electro type electrode type because here the activity of hide the partial pressure of hydrogen is 8 atmosphere.
- And here the partial pressure of hydrogen at the cathode is 2 atmosphere.
- We know from the expression E cell is equal to 0.0591 by n log of partial pressure of hydrogen at the anode by partial pressure of hydrogen at the cathode.

Example or application from the lecture

- So the first question is we have been given a cell where we have coupled the iron electrode with the silver electrode.
- What is given to us is the.
- Let us find out the E naught cell, which is given by E naught of the cathode minus E0 of the anode.

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EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 05:00–09:59

At the cathode, this is the reaction.

Key details

- At the anode, the reverse reaction would be taking place.
- H2 is giving you 2H plus 2 electron that is at the anode.
- And the cathode partial pressure would be 2 atmosphere.
- When we find out the cell potential expression, it comes out as 0.0591 by n log of partial pressure of hydrogen at anode by partial pressure of hydrogen at the cathode.
- And understand here E0 is not there because E0 is same for the anode as well as the cathode and hence cancels out.
- And the answer now is 0.1779 volts.
- What do you understand by decinormal calomel electrode?

- We know that the.
- Potential of the calomar electrode depends on the chloride ion concentration or the KCl concentration which is used.
- A saturated calomel electrode.
- That forms the anode.
- We are asked to write the cell representation and calculate the concentration of Cl minus IN the saturated calomel electrode.
- Please understand that we already know that the cathode concentration of KCl is 0.1 molar or normal.
- And we need to determine the chloride ion concentration for the saturated calomel electrode.
- Anode, it is a saturated calamar electrode for which the concentration is not known.
- The decinormal calomel electrode, and for any calomal electrode, we know the expression is E naught.
- Minus 0.0591 log of the concentration of Cl minus.
- Now here looking at the cathode, it's a decinormal calomyl electrode, therefore Cl minus is 0.1, and here it is.
- X because we want we don't know what is the concentration of Cl minus.
- Calomyl electrode when it is a decenormal calomal electrode, the Nernst equation for the decenormal calomal electrode minus the Nernst equation for the saturated calomel electrode.
- Therefore, when we substitute we get 0.0988 is the e cell divided by 0.0591 log of x by point one.
- And therefore, when we substitute the values.
- We get we need to take antilog over here.
- When we do the antilog, we find that x is equal to 4.69 molar.
- Come to the next problem now.
- Here we have silver chloride 0.1 molar concentration of chloride ion.
- Can we identify this is a silver chloride electrode.
- Which is a DC normal or having a concentration of 0.1 normal or 0.1 molar chloride concentration.
- KCl concentration or HCl concentration is 0.1 molar.
- And the other side we have Fe two plus.
- Which has a concentration 0.29 molar, Fe 3 plus, which has a concentration 0.18 molar, and this is in contact with an inert electrode platinum.
- If you recognize what is this type of electrode, where we have.

Example or application from the lecture

- Let us look at a third type of problem where we have a decinormal calomyl electrode.
- First, let us write the cell representation.

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EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 10:00–14:59

Both the oxidized and the reduced species present together, this is a redox electrode.

Key details

- It says write the half cell reactions and overall cell reaction.
- And calculate E naught cell and E cell at 298 Kelvin.
- What is the solution?
- So first we need to write the half cell reactions.
- What would be the oxidation reaction for silver chloride electrode?
- We have already learned it.
- We will have the reaction Fe3 plus taking up electron and becoming Fe2 plus.
- That is the reduction reaction.
- Whenever we have such a question.
- It's always a good practice to write the half cell reactions and then the overall cell reaction, even though it is the question just says write the overall cell reaction.
- And next, the first thing we should do now is to find out what is Q because this has to be done very carefully.
- F E two plus by Cl minus.
- Fe three plus.
- Cl minus and Fe 3 plus.
- That is the value of Q.
- Yes, that would be E naught cathode minus E0 anode.
- We got E naught cell.
- And N I know is 1 because there's only one electron getting exchanged in the overall reaction also.
- And now I have an expression here that is the Q expression.
- How did I get this?
- This I got from the overall cell reaction.
- We are the product concentration divided by the.
- Reactant concentration and therefore we find if we substitute 0.548 is the E naught cell and for Q we substitute the values 0.29, 0.18 and 0.1.
- And we find that.
- The overall cell E cell is 0.4767 volts.
- So that means we are straight away looking at a concentration cell.
- This is now we can make out the electrolyte concentrations are different.
- Therefore, this is an electrolyte type of concentration cell.
- Three plus three electrons is giving you.
- AU and this will happen at the cathode and at the anode we have the reverse that is AU is giving you AU3 plus three electrons.
- It is three.
- Therefore, if we substitute three here.
- And we know the concentrations at the cathode and the anode, we find that since n is equal to 3, E cell comes out to be 0.00749 volts.
- What you have to understand here is that whenever we do concentration cells as explained.

Example or application from the lecture

- Given the standard reduction potential values of both the electrodes are given to us, R is 8.314 joules per Kelvin per mole per mole and f is equal to nine six five zero faraday is nine six five zero coulombs per mole.
- Let us see.
- The values are given to us.

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EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

Core explanation · 15:00–19:59

During the when I was talking about concentration cells teaching you, e cell will be a very small value.

Key details

- You can see here E cell is a very small value, that's because the main contribution to e cell comes from E naught cell and E0 cell.
- From the cathode and anode being same as E naught of the anode and cathode being same, so E0 cell comes out to be zero.
- E0 for anode and cathode, they've got cancelled, therefore there is no E0 cell at all, and hence the concentration is.
- The cell potential is 0.00749 volts.
- That means it's a very small value.
- And since both the oxidized and the reduced species are present together in the same solution and an inert wire is placed to collect the potential, this is a redox electrode.
- We have to write the cell representation, write the reactions at the electrodes, and we have to calculate E naught cell and E cell at 298 Kelvin.
- Here we have to write the cell representation.
- Therefore this we place at the anode and this will be the cathode.
- The reactions at anode would be C U plus is giving you CU2 plus 2 electron and here at the.
- Cathode we have this reaction, the calom electrode is getting reduced.
- Hg2Cl2 plus 2 electron is giving you 2 H G plus 2 Cl minus.
- Overall reaction would be 2CU plus H2Cl2 is giving you 2 C U2 plus 2 Hg plus 2 Cl minus.
- E0 cell will be E cathode, E0 of the cathode minus E0 of the anode, that is 0.281 minus 0.153, and the value for E0 cell is 0.128 volts.
- E no E cell.
- For E cell I need Q.
- So first itself, so C U2 plus using the overall reaction.
- C U2 plus squared.
- And here it is Cl minus.
- squared and then that is divided by.
- C u plus square.
- And so we substitute that expression over here.
- And n in this case we can see is two electron.
- of C U2 plus Cl minus and C U plus and these are substituted here and N is equal to 2.

- We have to write the E naught cell and E cell at 298 Kelvin.
- E naught for the gold electrode is 1.52 volts and E naught.
- For an iron electrode is minus 0.44 volts.
- F is nine six five zero coulombs per mole.
- We know the half cell reactions that is Fe iron is undergoing oxidation, so Fe 2 plus 2 electron, and here we have AU3 plus 3 electron at the cathode undergoing reduction A giving you AU.

Example or application from the lecture

- Let us look at the next problem where we have a decommal calomel electrode and it is used to determine the potential of the following redox electrode.
- So given the standard reduction potential for the calomar electrode that means that the chloride concentration is 1 molar and therefore this is 0.281 volts and C U2 plus C U, the standard reduction potential is given as 0.153.
- Let us look at the solution.

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EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 20:00–24:59

That's exactly what we have done here, and the overall reaction now we get 3 FE plus 2AU3 plus, giving you 3 Fe plus.

Key details

- 2 plus 2 AU.
- That is the net cell reaction.
- As I mentioned before, always good to write the Q first before proceeding with the calculation.
- If it was not at 298 Kelvin, then we would require these values.
- Because in that case, what is the expression you would use?
- 2.303 R T over NF.
- By AU plus 3 squared.
- That's exactly the Q value here.
- What is N now here?
- has this will become 6, this will also become 6, therefore the number of electrons are going to cancel each other.
- And then we get n as 6.
- And when you substitute the substitute the values, this is the expression we get and the final answer we get it as.
- One point nine eight three six votes.
- which is not known to us.
- The cell potentials are 0.215 volts and 0.385 volts, which are when the electrode is in contact with the solution of pH is equal to 7 and with the solution of unknown pH.
- E cell plus 0.0591 PH plus E saturated calomol electrode.

- So this expression first will help us to determine E naught G because E naught G value is not known to us unless we use.
- A solution of known pH.
- What is the pH here?
- So first and foremost we find out what is E0 G, and that is 0.8727 volts, and then we use this expression.
- P H is equal to E naught G minus E SE minus E cell divided by 0.0591.
- And E saturated calomin electrode is known to us, and E cell measured.
- Now we have to use this value because now the pH is unknown.
- Here, what is the E cell we use?
- Because this is the pH seven, we know the pH seven.
- T cell we are going to use now.
- When we do that, we will substitute these values and what we get is ph is equal to 4.12.
- Module of electrochemical equilibria.
- Now in the next class we will start with unit three, that is on energy storage devices.

Example or application from the lecture

- And let us write the Q again.
- Now let us find out the E cell as that is what we is to be determined E cell at 298 Kelvin, therefore 0.0591.
- Let us come to the last problem.

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EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

8 Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1

Study focus: Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000130 · EV0000131

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module six on corrosion chemistry and today we will discuss the types of electrochemical corrosion, that is differential metal corrosion and differential aeration corrosion.

Key details

- When we say types of corrosion, we have studied that there are two types.
- One is dry corrosion and the other one is wet corrosion.

- Dry corrosion, as we understand, is due to the chemical attack of the environment on the metal surface and it happens in the absence of any conducting medium.
- But for wet corrosion it has to take place only in conducting medium because what we have is an electrochemical attack.
- And since most of the times metals are exposed to conducting medium because there's always moisture in the air, therefore wet corrosion is of more importance to us.
- When we have two.
- Dissimilar metals in contact with each other and they are exposed to the corrosive environment, that is when we say galvanic corrosion will take place or differential metal corrosion.
- Two different metals in contact with each other.
- Therefore, galvanic corrosion can happen very frequently in metals.
- When we have two different metals.
- They will have different electrode potentials.
- The moment they are in contact with each other, one of them which has a greater oxidation tendency becomes the anode, and the one which has lesser oxidation tendency will remain as the cathode.
- Now, because there will always be in a change in the electrode potentials, this situation or formation of anode and cathode can take place very easily.
- Now because we are we are talking of electrochemical attack, the most important thing which we should see under any situation is how are this anode and cathode getting formed on the surface and in the presence of the medium conducting medium we will have corrosion happening.
- Now here because one of the metal has a lower electrode potential or rather.
- Negative electrode potential that becomes the anode, the other automatically becomes a cathode.
- And we know that it is only the anode which undergoes oxidation and hence gets corroded, whereas the cathode, the metal which has become cathode, will remain unaffected by carrot.
- More the difference in the electrode potentials of the two metals.
- More is going to be the corrosion which will happen, of course, it is only the anode which will corrode.
- in the presence of oxygen and four electrons giving you four OH minus.
- We learned that at the cathode there can be different reactions based on the medium.
- In the atmosphere, and there is moisture.
- So, therefore, this is the most common reaction which can happen at the cathode.
- Like we have iron in contact with copper.
- It is minus 0.44, and copper is 0.34.
- Copper will become the cathode.
- The metal will get converted into its ions, therefore the metal is getting destroyed.
- Whereas at the cathode we know that copper cannot undergo reduction.

Example or application from the lecture

- Let us understand what is differential metal corrosion.
- Let us see what will be the reactions.

- Now we'll consider this most commonly happening because there's always plenty of oxygen.

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EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 05:00–09:59

Unaffected by this type of corrosion when it is in contact with iron.

Key details

- And what will be the reaction which is happening?
- Just now we said that oxygen with the H2O plus two electron is giving you 2 OH minus.
- Earlier it was iron in contact with copper.
- Therefore, what will happen?
- Zinc is a more active metal as compared to iron.
- Therefore, zinc will now act as the anode and the cathode will now be the iron.
- Iron was undergoing corrosion, whereas when it is in contact with zinc, it is zinc which will undergo corrosion.
- In the first case, where we had iron in contact with copper, we will get copper hydroxide iron hydroxide.
- Very common is when we have steel screws or rivets in copper sheets.
- That it we observe here in this case, in this picture, that it is the steel nut which is getting corroded whereas the bolt remains absolutely intact.
- And similarly if we have a copper sheet with steel screws, there also we will find that the copper sheet remains in sack intact and it's only the steel screws which will get corroded.
- And very common again you can see this nice picture which is showing the steel pipe connected to copper plumbing.
- But what is getting corroded?
- It is only the steel pipe and not the copper plumbing.
- this is very commonly seen if you have an iron rod which is lying, you can have a part of it which is exposed to different types of concentration of the oxygen could be different in one area and it could be different in the other area.
- Which is exposed to lower concentration of oxygen becomes anodic in nature and undergoes corrosion, whereas the other part of the metal which has a higher oxygen concentration becomes cathodic and is unaffected by the.
- Now, the because of the difference in this oxygen concentration, we are getting something like a oxygen concentration cell on the surface of the metal.
- And this potential difference will cause the current to a small amount of current corrosion current to flow.
- We have anode the same metal will undergo corrosion.

Example or application from the lecture

- Another example, let us take where we have now iron in contact with zinc.
- So this is a very clear example of galvanic corrosion or differential metal corrosion.
- Let us see what are the reactions and you can see here.

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EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

Core explanation · 10:00–14:59

Oxygen absorption is the reaction we are going to look at.

Key details

- We have one part which is.
- Immersed in water or is less oxygenated, and the other part is more oxygenated.
- It could be water, it could be any other area which is less oxygenated.
- One part is exposed to less oxygenated, another part is exposed to more oxygenated region so that is exposed to more oxygen.
- Now because the oxygen absorption takes place at the cathole, so higher concentration of oxygen wherever we have.
- That area becomes the cathode, whereas the other part where there is less oxygen now becomes the anode, and because it's the anode which undergoes corrosion, that is the metal will take part in this reaction here, whereas here there's a lot of oxygen, so oxygen will be involved.
- the is always involved at the cathodic region so this area exposed to more oxygen is going to become the cathode whereas the part exposed to less oxygen is going to become the anode.
- When you pull out a nail from the wall, what do you observe?
- The part of the nail which is inside the wall, you can see here the part which was inside the wall.
- The wall has got rusted, whereas the part which was outside did not get rusted.
- The reason that because this part was exposed to less oxygen, that this acted as the cathode and the part exposed to.
- Sorry, this part which was exposed to less oxygen became the anode and the part exposed to more oxygen now has become the cathode.
- We see in this case that the part exposed to.
- Exposed to less oxygen becomes the anode and the part exposed to more oxygen becomes the cathode.
- And therefore we see the formation of anode and cathode on the surface of the nail and the part inside the wall which was exposed to less oxygen is getting the same.
- getting corroded whereas the part exposed to more oxygen is not getting corroded.
- As you can see in the picture, the part where they are crossing each other, the two wires are crossing each other, you can see that it has got corroded, whereas the part exposed outside it is not getting corroded.
- The reason again the part which is crossing each other there the amount of oxygen is less and therefore that acts as the anode whereas the part exposed to more oxygen now this other part so.
- These parts you can see are more corroded.
- You can see the yellowish orangish color over here, the rust getting formed, whereas the other part look intact.
- That is acting as the cathode.

- That part is getting corroded, whereas the part exposed to more oxygen is not getting corroded.
- What do we see there?
- The part which is inside the paper has got corroded.
- All of you would have experienced this.
- The part which is inside the paper has got corroded, whereas the part which was outside of the paper has not got corroded.
- Now in tomorrow's class, in the next class, we are going to talk about a special type of differential aeration corrosion.
- And that is waterline corrosion.
- And we also go to another type of corrosion that is pitting corrosion.

Example or application from the lecture

- And let us assume that it's a neutral medium.
- This is a wonderful example.
- Another good example would be a wire mesh.

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EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

9 Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II

Study focus: Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000142 · EV0000143

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module six on corrosion chemistry and we will talk about different types of electrochemical corrosion and today's discussion will be on waterline corrosion and petting corrosion.

Key details

- Last class we discussed about differential aeration corrosion where an oxygen concentration cell was getting formed due to difference in the concentration of oxygen to which the metal surface is exposed.

- This happens when a metal is half immersed in water.
- And when this happens we call it as waterline corrosion.
- Whereas at the cathode because there is more oxygen.
- And if you remember electrochemical theory of corrosion, I said that the.
- Products which are formed at the anode and cathode, the ions which are formed at the anode and cathode move towards each other and in between the anode and cathode we get the formation of the corrosion product.
- The corrosion product will deposit in between the anode and cathode.
- And that's exactly what happens here.
- A brown line on the surface of the ship.
- Many of you would have noticed this, or there will be a brown line on the surface of the ship, and where do you see it?
- Wherever the water line.
- Brown line and what is that brown line on the surface?
- It is due to the corrosion product.
- they move towards each other and somewhere in between they will get deposited and that is happening at the water level.
- Which is formed just below the waterline, and because of this, this type of corrosion is referred to as waterline corrosion.
- So basically it is differential aeration, but because it is happening in the presence of water, especially, it is called waterline correction.
- Corrosion and the brown line appears at the waterline.
- What happens here?
- The part which is inside the inside the water now here again exposed to less oxygen that becomes the anode and the part which is above exposed to more oxygen is going to become the cathodic area and in between we will see a brown line getting.

Example or application from the lecture

- Let us see, take up some examples and understand this.
- Now let us take up another example where we have a steel tank which is partially filled with water and left for a long time.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

Core explanation · 05:00–09:59

formed on the surface of this steel tank.

Key details

- So half of it is inside the water, again the same reason we have less oxygen, therefore iodic area is this oxy this part.
- And the more oxygen part would be the cathodic area, and somewhere at the waterline we will see the formation of the corrosion product.
- And now, since there is sodium chloride, we know that.

- The conductivity of the solution will be very high, and therefore we find corrosion also will be more in this case when we have a solution.
- Of course, it is aerated, so obviously there will be oxygen in it, but never.
- the less oxygen.
- What happens here?
- It is a localized and accelerated corrosion, and because of this, there is formation of pits and pin holes around which the metal is relatively unattacked.
- You can see in this picture how pits have got formed on the metal surface, and it's only this area which has got corroded, whereas the rest of it looks good.
- And therefore we find that formation of pits and pinholes.
- Therefore it is called pitting corrosion and the rest of the area of the metal looks intact.
- Because if there are very small pinholes getting formed, the corrosion product itself is going to cover the pinholes and we will not even know till the destruction has happened.
- Which is carrying oil or water.
- And the pipe will suddenly start till it is completely damaged, we will not come to know and unless it starts leaking, it will just give out.
- the water or the oil and that is where that is when we come to know but the damage is already done.
- So what is the characteristic very important here for us to understand it is characterized by small anode area and a large cathode area because the anode area is very small it will result in accelerated corrosion.
- Now, because the water droplets is covering a small part of the metal surface, the entire surface is now not exposed to the atmosphere, a small part.
- That is covered by this water droplet.
- As a result, what is going to happen?
- You can see there is a small anode and a large cathode, as I said just now that a small anode and large cathode is characteristic of fitting corrosion, and we will find that the anode.

Example or application from the lecture

- Another example would be when an iron rod is partially dipped in a sodium chloride solution.
- We have to understand that waterline corrosion is also differential aeration corrosion, but only a special case because we consider these cases wherever the metal is partially immersed into the water.
- For example, if this is a pipeline.

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EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 10:00–14:59

Anode being small undergoes corrosion at a very fast rate.

Key details

- You can see again here there's a small anode and there's a large cathode which is there.

- Now because the area the cathode is so large, the demand for electrons is very high.
- from the cathode.
- The anode will undergo corrosion at a faster rate.
- And because the anode is very small, we will find that it undergoes corrosion at a very fast rate being small there is like a.
- Which is getting formed in this anode area.
- Why accelerated corrosion?
- We can understand because the demand for electrons is very high, the cathode being large.
- Taking part in the reaction, of course, here because it is exposed to more oxygen, we will say this is the reaction which is happening under the dust particle or the water particle, it is anode, so it is the metal.
- Say we were talking of iron or steel, so iron will become Fe2 plus.
- There is destruction, and this pit will go on deepening slowly till it reaches the end.
- And till that time we will not be able to even discover that there is a pit, and that will lead to a damage of the structure.
- could be of fitting corrosion where we have this protective film which got or formed on the surface which is be been made on the surface has got damaged.
- This process is calling.
- But the moment there is a damage or there's a small hole or it got scraped, then this is going to expose the iron metal to the surface.
- And this is minus zero point four.
- These are the electrode potential values.
- Iron is going to act as the anode and tin is going to act as the cathode.
- So this now will result in the anode undergoing corrosion at a very fast rate and therefore since the anode is very small it will be like digging a pit in this area.
- So therefore, we find that in this case tin is acting as the cathode.
- The value is minus 0.18 or maybe 0.14.
- And here it is 0.44, and therefore we find that this metal.
- The steel is going to act as anode, whereas the tin is going to act as cathode, and now anode being very small, the demand for electrons from the cathode, the tin coating is the cathode large cathode, demand for electrons is very high, and therefore it forces the anode.
- To undergo corrosion at a very fast rate, and because the anode is small, it will be a pit will be dug here, and then this is how the damage is going to happen.
- Or we say galvanic corrosion, whereas in the case one we saw it was differential aeration corrosion.
- But what is common to go?
- That they have a small anode and a large cathode, and that is what is characteristic of pitting corrosion.

Example or application from the lecture

- Let us understand this better.
- For example, this is a very common thing which we see iron being covered by tin.

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EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 15:00–19:59

Tin is not going to be involved in the reduction reaction, it is only the environment, and here we have oxygen.

Key details

- So therefore this is the reaction which will take place.
- Hence the anode will get corroded and a deep pit, narrow pit will be formed at this position.
- What we have to remember small anode and large cathode.
- Whenever we have this situation, it is going to result in pitting corrosion.
- So before that let me tell you that in the next class we are going to discuss the last type of corrosion that is stress corrosion.

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EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

10 Types of corrosion - stress corrosion - Part III

Study focus: Types of corrosion - stress corrosion - Part III. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module six on corrosion chemistry.

Key details

- Then we move on to studying some of the factors which affect the rate of corrosion.
- Stress corrosion is also referred to as stress corrosion cracking, and this is a most common type of corrosion which happens with metals when they are under stress.
- and at the same time they are exposed to a specific corrosive environment.
- We know that whenever.
- We are doing any manufacturing or fabrication of articles, structures, etc., the metal is subjected to lot of mechanical operations such as pressing, hammering, rolling, bending, etc.
- Now, because of these operations on the metal to get the right structure there will be some a residual stress left inside the metal.
- Apart from that, that there could also be some external stress due to the service conditions.
- Give the right corrosive environment.
- We have an iron article.

- Is so much here, then a small crack is going to be initiated for inside the metal in order to relieve the stress.
- the bend which is here, the bent article over here.
- Now this being at higher energy level, they will be more reactive as compared to this part, and therefore the reactive part will become the anode and the stress free part becomes the cathode.
- We have to see how the anode and cathode are getting generated, and in this stress corrosion, this is how it is being formed wherever there is stress in the metal.
- Exposed to a specific corrosive environment, we can see that stress corrosion cracking is going to take place.
- We just now saw that there could be tensile stress in the metal but apart from that the specific corrosive environment is what is required.
- Just having the specific corrosive environment or the tensile strength is tr stress is not.
- Yeah sufficient both these factors the tensile stress as well as the specific corrosive environment when they act upon this metal, then the metal undergoes stress corrosion.
- The stressed part and the other part is acting as the cathode.
- This crack when we subject it to a corrosive environment we will find that due to corrosion this part being anode undergoes corrosion and we know what happens in corrosion the metal is going to get converted into iron.
- It's going to deepen and by the time it reaches here it has caused damage or breakdown of the structure.

Example or application from the lecture

- First, let us look at what is stress corrosion.
- Suppose we have a metal which is bent like this and a small craitiated in it.

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EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

But what we have to remember is both the conditions should be satisfied.

Key details

- The tensile stress as well as specific corrosive environment needs to be present for such a corrosion to take place.
- If the metal is break brass or the structure is made of brass, which is an alloy, this will be corroded if there is stress in the structure.
- This will get corroded.
- only in the presence of ammoniacal solution or ammoniacal vapor.
- cracking to take place.
- So therefore in the presence of both these conditions we will find that stress corrosion can happen and one more.
- Fact about stress corrosion is that it causes serious damage as we saw this crack once it starts deepening due to the corrosive environment, it can cause breakdown of structures and

it has in the past causes caused air crashes, bridge collapses, and even border.
- Just now I mentioned that boiler explosions could happen due to this stress corrosion.
- Basically it's a form of stress corrosion and it takes place in boilers and we know that in what boilers what are we doing?
- We are heating water.
- If this comes in constant contact with concentrated alkali, then there is stress in the boiler and also you provided the specific corrosive environment, stress corrosion is going to happen.
- we get sodium hydroxide.
- And if we remember that for mild steel, what is the corrosive environment we require sodium hydroxide.
- there now what happens is because we are heating the boiler, it is at a high temperature, this sodium hydroxide which went inside the crack.
- is slowly getting more and more concentrated.
- When it gets concentrated to about 10%, then this sodium hydroxide starts dissolving all the iron with which it is in contact.

Example or application from the lecture

- Let us talk about it.
- Let us talk of how distress corrosion takes place in boilers.
- Let us see what exactly is happening.

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EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

Core explanation · 10:00–14:59

And when it dissolves it forms a compound called sodium ferroate.

Key details

- After this, the sodium ferroate undergoes further hydrolysis, and when it undergoes further hydrolysis, it is going to look at this reaction sodium fydrohy ferro weight is undergoing hydrolysis giving you Fe three O four plus H two plus NOH.
- Again you have regenerated the sodium hydroxide.
- So that means throughout the you while you are using the boiler continuously you have sodium hydroxide in the water and apart from that we also got Fe three O four and Fe three O four is known to be brittle in nature.
- The iron which was around this crack which was initially gener initiated due to the due to the stress in the structure.
- Got dissolved and now it is forming something like Fe 3O4, which is brittle in nature.
- The crack now starts deepening.
- Because it is happening in the presence of sodium hydrogen.
- hydroxide and it is in a very specific environment.
- the mild steel is the water boiler made of mild steel is undergoing corrosion.
- Plus we are making the iron into something brittle like FE3O4 I mentioned just now.
- Therefore, this is called as this is known as caustic embrittlement.

- So because of this causing embrittlement, we get a corrosion cell.
- Whenever we talk of electrochemical corrosion, we know there should be a formation of anode and cathode.
- We'll always follow the same rule.
- We'll write the anode first and then the cathode.
- And it is in contact with concentrated sodium hydrox, whereas the stress free iron is going to act as the cathode, and that is in contact with the dilute sodium hydroxide.
- And that is the reason why electrochemical corrosion takes place, the stress as well as the specific environment in this case we have provided the sodium hydroxide.
- Now of course we have to think of ways by which this can be prevented because it could be very dangerous.
- Then we have the difference in electrode potentials of anode and cathode, nature of corrosion product, ratio of the anode to cathode area, then we have overvoltage.
- Polarization, pH, temperature, conductivity of the medium and humidity.
- The first one, nature of the metal.
- How much corrosion is going to happen is going to depend on how reactive the metal is.
- If it is a noble metal, we don't expect much reactivity.
- Therefore corrosion is also very limited.
- If it's very noble, then you hardly see any corrosion.
- Those metals which are very reactive we find that they will be undergoing corrosion.
- In fact, they are more susceptible to corrosion.
- So we can say that the ones with lower electrode potential, that is where the reduction potentials are negative, that means they want to undergo oxidation, and we know that corrosion is an oxidation process.

Example or application from the lecture

- Let us take up some of them.

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EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 15:00–19:59

Negative reduction potentials will be more reactive than those which have positive reduction potential values.

Key details

- They all have very negative values, and we know that they are highly susceptible to corrosion.
- On the other hand, we have noble metals like silver, gold, and platinum which have positive values and we find that they are less susceptible to.
- And that's exactly what we found the ones which are lying high above in the series will undergo corrosion.
- The ones which are very low in the series with positive values do not undergo corrosion.
- But there are exceptions like chromium, aluminium, titanium.

- Because they become passive.
- Now that means electrochemical series is not sufficient for us to give us the order in which corrosion will happen.
- Therefore, a new series is required, which is known as the galvanic series.
- And what does this galvanic series do?
- It is going to arrange the metal.
- And now we have an advantage, it also can arrange the alloys in the order of their tendency to undergo corrosion and in a particular environment because we know that corrosion depends not only on the metal or the alloy, it also depends on the environment to which the metal is subjected.
- Yes, to a good extent we can say that the electrochemical series gives us an order in which the metals will show tendency to undergo corrosion, but nevertheless, electrochemical series are standard values.
- That means we are using standard solutions and standard conditions, and moreover.
- Aluminium, you see here it has a very negative value.
- Aluminium, chromium also you'll see they have very negative values, but we know that aluminium is passive in nature.
- So that means electrochemical series is not sufficient.
- For us to decide the corrosion tendency.
- Tend and see not only metals, you can see some alloys also.
- All these are arranged in an order in which.
- They will undergo corrosion.
- and listed here which metal or alloy will be more reactive or undergo corrosion more as compared to the others.
- Always the environment is also listed over there.
- Coming to the next.
- reason or the factor which is going to affect the rate of corrosion and here the difference between the potential of the anode and the cathode region.
- We know that many times two metals are in contact with each other and.
- This is say M1 and this is M2.

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EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 20:00–24:59

We know also that the driving force here is the difference in the electrode potential.

Key details

- We are talking about galvanic corrosion.
- Higher is going to be this differential metal corrosion or galvanic corrosion.
- Because more is the potential difference, diff more is the potential difference, the free energy decrease accompanying the process is higher, and therefore we find that extent of corrosion also increases.

- But what we have to remember here, more the difference between the electrode potentials of the two metals, more is the anode which is going to undergo corrosion.
- It is a more reactive metal which undergoes corrosion.
- If we understand that this is the anode and this is the cathode, we will find more the difference between this anode and cathode.
- More is the corrosion of M1 metal in this case.
- So what is the first of all which metal will undergo corrosion?
- Iron, we can see very clearly.
- Because this will become the anode.
- And the potential difference here comes out to be 0.78 volts.
- What is the metal which is undergoing corrosion?
- Again we can see that it is iron which will undergo corrosion.
- Here also I am acting as anode, but now the potential difference has come down to 0.3 that is e cathode e0 cathode minus the e0 anode.
- Here it is 0.78, here it is 0.3.
- Therefore, though in both the cases it is iron which.
- We find that iron will undergo corrosion faster when it is in contact with copper.
- The use of this similar metal should be avoided wherever it is possible.
- Should be always made of the same metal, otherwise, we'll find that the metal which is more reactive.
- Will undergo corrosion.
- Because nuts and bolts are always in close contact, they should be made of the same metal.
- Now coming to the next very important factor which affects the rate of corrosion, nature of corrosion product.
- Now in some cases, just now we saw in the first factor that cr when we talked about nature of metal, chromium, aluminium, etc., they do not undergo corrosion.
- They become passive.
- Because once the corrosion product is formed on the surface of the metal, it acts as a protective film.
- This we should always remember.
- The metal, and now the corrosion product has got formed on the surface.
- product which is formed on the surface.
- So it's forming a nice film over the surface and protecting the metal how it is creating a barrier between the surface and the environment and therefore we will find that the metal is now cut off from the environment and hence there can be no further corrosion.
- So, if the corrosion product is insoluble, stable, uniform, and non-porous, we find that it forms a protective film and therefore prevents further corrosion.
- It acts as a barrier, a thin, invisible, impervious sometimes continuous film is formed, and this acts as a barrier between the fresh metal surface and the corrosion environment.
- Passive, and that's exactly what we see in the case of chromium and aluminium.

Example or application from the lecture

- Let us take up a case one where we have iron which is minus 0.44 volts and copper 0.34 volts.

- Let us take another case where we have iron at minus 0.44 and tin is the metal which is in close contact with it and that tin has an electrode potential standard electrode potential of minus 0.14.
- For example, nuts and bolts.

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EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 25:00–29:59

metals like iron, zinc and magnesium.

Key details

- A non-continuous film will be formed.
- Even though the corrosion product is formed, it is not protecting the surface underneath, and every time there is a new surface after the corrosion product is formed, there's a new surface which is getting exposed.
- and therefore the corrosion will continue to happen continuously.
- So metals like iron, zinc, and magnesium, the corrosion product is soluble, unstable, non-uniform, and porous, and therefore there is no stopping corrosion.
- It will keep on getting corroded as a new metal surface, a fresh metal surface is getting extra.
- Which is covering the surface, but it is soluble in the surrounding environment.
- getting exposed every time.
- that this will result in more and more corrosion of the metal.
- We all use stainless steel utensils for cooking and other work in the at homes.
- But we know that it does not get corroded.
- Because it has.
- It is an alloy which contains chromium.
- steel or that is iron it iron containing some amount of carbon so that steel articles we find that they get corroded easily like your nails or any steel article will get corroded but stainless steel due to the.
- Passive nature of chromium which has been added there.
- This chromium oxide forms a nice layer on the surface and forms a protective film and prevents further corrosion of the particle.
- Now in tomorrow's class in the next class we are going to talk about take up some more factors like overvoltage, polarization, pH temperature, etc, which were listed.
- It just now, and then we will learn how those factors are going to affect corrosion rate.

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EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

11 Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage)

Study focus: Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000182 · EV0000183

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module six on corrosion chemistry, and today we are going to study about some more factors affecting the rate of corrosion.

Key details

- The size of the anode and the cathode plays a major role in deciding the rate at which the anode undergoes corrosion.
- We remember from electrochemical theory that it's only the anode which undergoes corrosion and the cathode remains unaffected.
- Anode area being very small and a cathode area being very large, corrosion is very intensive and the anode will undergo corrosion at a very fast rate and as this ratio becomes smaller and smaller the corrosion rate is going to increase rapidly and this is what we studied under pitting corrosion.
- Is the reason for that.
- We know that when the anode is small, as in this case, and the cathode is large, the electrons are getting liberated at the anode and they are getting consumed at the cathode and since the cathode is so large there is rapid consumption of these electrons and the cathode and like and because of that demand for electrons is very high from the cathode.
- Therefore the anode has to undergo reaction at a very fast rate.
- That means it will get corroded at a very fast rate.
- If the tin coating is broken due to some reason, then we find that iron surface is getting exposed to the environment, and therefore, because iron is anode, it will be iron which will undergo corrosion.
- And since we have a case of small anode and large case.
- Whereas if we have covered iron with zinc.
- The next factor is about hydrogen over voltage.
- Overvoltage is defined as the extra voltage which we need to provide over the theoretical voltage for a particular electrode reaction to take place.
- But we find that the reaction does not take place specially if there is evolution of gases.
- It is noticed that.
- In the case like say hydrogen gas or oxygen gas is getting evolved at the electrode due to the reaction, it is found that we need to provide this extra voltage, and that voltage is referred to as the overvoltage.
- And this overvoltage value depends on the.

Example or application from the lecture

- Let us study the ratio of anodic to cathodic area.
- For example, let us say we have an iron article which is covered with tin.
- So let us first understand what is overvoltage.

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EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 05:00–09:59

Hydrogen over voltage values and also ox oxygen over voltage values.

Key details

- When we have just platinum, we have a value which is close to zero point one.
- But if you take the case of platinized platinum it's almost going to zero.
- That means the same hydrogen evolution on a platinum surface would not be would be more facilitated when it's platinized platinum why because the surface area as has become high so in the case of platinum we find that platinized platinum has lower hydrogen over voltage so what would be the result the hydrogen gas evolution which we will be much higher.
- This is the hydrogen voltage values.
- And then coming to platinum, it is even more easier.
- to for the hydrogen liberation and when it comes to platinized platinum it's even more the hydrogen over voltage is very less.
- This shows that hydrogen over voltage is dependent on the surface of the metal on which the hydrogen evolution is taking place.
- If a metal has low hydrogen overvoltage on its surface, then it is more susceptible to corrosion.
- Why would we say that?
- Because hydrogen liberation H plus electron is giving you half H2 is a cathodic reaction.
- Now, if in a particular during a particular corrosion, if cathodic reaction is hydrogen evolution type, then we will find that hydrogen gas gets liberated easily, and that means.
- The cathodic reaction is very fast.
- Now if the cathodic reaction is very fast, it is going to make the anodic reaction also fast because we know that anodic and cathodic reactions will take place simultaneously, and if the anode is undergoing reaction at a fast rate, it means that the high.
- The over overall corrosion is going to happen at a fast rate.
- And if the.
- Hydrogen overvoltage on the metal surface is high, that means the cathodic reaction has become slow and therefore the oxidation also becomes less and therefore the reac the corrosion of that metal will also become slower.
- We know that for any chemical reaction, as temperature increases, there is going to be an increase in the rate of reaction.
- And similarly, that is what we observe in the case of metals also or in the case of corrosion process.
- What are the various reasons?

- Apart from this, there are several other reasons which are going to promote corrosion due to high temperature.
- conductivity of the ionic conductance will increase with temperature.
- I have mentioned before, whether it's in batteries or in corrosion, it's not just the anode and cathode reaction which will decide the fate of the reaction.
- It's also the movement of ions in the electrolyte.
- the reaction will also be faster.
- We will study about this polarization soon and we will understand what is this polarization at the anode and cathode sites.

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EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

Core explanation · 10:00–14:59

the because of high temperature polarization effect is less and therefore we find that corrosion will be faster.

Key details

- Also the s corrosion product solubility of the corrosion product generally increases at with high temperature.
- Solubility of any substance we know with increase in temperature it is going to increase.
- So here because this for the product is on the surface and if it is getting more dissolved into the solution into the medium we can understand that it's going to expose new surface every time and therefore we find that the rate of corrosion is also going to increase and apart from that another reason could be breakdown of the protective film.
- Now we know that many times an ice protective film gets formed over the surface but at a higher temperature it is possible that this the covering is going to the protective film is going to break down and therefore that could be another.
- Reason for height as the rate of reaction being higher or rate of corrosion being higher at a higher temperature.
- But there is an exception which we should know that oxygens like sorry, gases like oxygen carbon dioxide, if they.
- They dissolve, they are and are causing corrosion, then the rate of corrosion decreases with increase in temperature as the solubility of these gases in the medium decreases.
- So if the temperature is more than oxygen carbon dioxide, solubility is less and therefore we find that the rate of corrosion actually decreases.
- We know in general that when the pH is low, we expect that the rate of corrosion will be high, meaning that in acidic medium we find that the rate of corrosion is higher.
- Why is it so?
- Because when we have low pH.
- we find that the concentration of H plus ions is very high and we know that H plus ions are responsible for one of the cathodic reaction.
- If this H plus ion concentration is more, obviously we expect that this reaction will take place at a very fast rate.

- Now if the if this reaction happens fast, this being a cathodic reaction, it is also going to make the anodic oxidation faster and therefore we expect that the rate of corrosion will increase.
- Apart from that another reason in acidic medium is that corrosion product becomes more soluble in acetic medium then.
- Every time again a new surface is getting exposed and therefore we find that the rate of corrosion increases.
- But there is an exception here which we have to note that metals like aluminium actually undergo corrosion faster in highly alkaline solution.
- Whenever there is a drift in the electrode potential of the electrode, it is known as polarized.
- We say that the electrode got polarized.
- If it happens at anode we say it's anode has got polarized.
- If it happens at cathode we say cathode has got polarized.
- Electrode potential one of the important reasons could be a change in concentration.
- We know that when concentration changes, the electrode potential is also going to change.
- This here, look at the picture here, we have a metal surface, and at the metal, at the anode, we have anode and cathode getting formed on the surface.
- Metal is forming.
- Mn plus n electrons.

Example or application from the lecture

- Because we know that the corrosion product, for example, in case of aluminium A L2O3 is more soluble in alkaline medium.
- So even if L2O3 is getting formed, because it dissolves in the alkaline medium, we find that for aluminium in alkaline medium the rate of corrosion is going to be more let us look at another important factor that is called polarization of the anode and the cathode.
- First, let us understand what is polarization.

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EV0000190 · EV0000191 · EV0000192 · +1 source units in pack.json

Core explanation · 15:00–19:59

But many times the rate at which the anode is undergoing oxidation and the rate at which the metal ions are moving towards the cathode is different.

Key details

- Metal ions is very high.
- In that case, we find there is an accumulation of metalions around the anode, and when this happens, obviously the tendency for this reaction to go in the forward direction will be reduced, and that means the tendency of the metal to undergo.
- Oxidation is reduced, or in other words, corrosion will be less now.
- So the rate of corrosion is less, and the anode is said to be polarized because there is a drift in the electrode potential due to accumulation of ions.

- What we have to understand that whenever anodic polarization happens, it is going to reduce the rate of corrosion because the tendency of the metal to undergo oxidation is actually decreasing, the reason there is accumulation of ions around the anodic region.
- I reduction of oxygen or reduction of H plus ions, H plus ions taking up electrons, oxygen taking up electrons.
- When we find that there is not enough oxygen and H plus which are close to the cat hole.
- We find that there is a depletion of oxygen and H plus around the cathode, and therefore there is no reactant to pick up these electrons and that would result in cathodic polarization.
- Because there are less number of oxygen or H plus ions around the cathode, the rate of cathodic reaction also decreases.
- And therefore, we say that the cathode has got polarized and again this is resulting in slowing down of the corrosion product.
- And also it could be due to retarded removal of the product.
- When the anode and cathode get polarized we see that the rate of corrosion comes down.
- But in certain cases, like in batteries, etc., we don't want polarization.
- One the types of depolarizers are anodic depolarizers, depolarizers, sorry, not polarizers, and cathodic depolarizers.
- Similarly, cathodic depulverizers like cuplicons, which are going to take up the electrons themselves, and therefore this results in.
- Another way of in removing the polarization effect is to increase the temperature.
- Just now while studying temperature we understood that depolarization, we talked about the polarization effect if it is there, then.
- Then it is with increase in temperature, the polarization effect is decreased.
- We can understood understand this because with increase in temperature, there will be a decrease in polarization.
- Because now the ions are more temperature, their ions can move easily, the migration of ions can happen and therefore polarization will be removed.
- Also simple things like stirring the solution.
- Basically you are.

Example or application from the lecture

- Let us talk of cathodic polarization.

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EV0000194 · EV0000195 · EV0000196 · +1 source units in pack.json

Core explanation · 20:00–24:59

removing the ions from the metal ions from the anode or you are causing the material which has to undergo reduction, the species which has to undergo reduction to come closer to the capsule.

Key details

- Next, electrical conductivity of the corrosion medium.
- Just now we have discussed it several times that.

- More is the conductivity, faster will the ions move between the anode and the cathode region of the corrosion cell.
- Now when this happens, we know that the exchange of electrons at the electrode surfaces is going to happen very quickly, very fast, and therefore the corrosion rate is going to increase.
- Saline water what does it mean?
- It contains salt.
- And s we know are good ionic conductors and therefore when there is lot of ions in this in the water then we find that they will migrate very fast and therefore the exchange of electrons will also happen at the ion and cathode very fast and therefore the rate of corrosion will always is be high in saline water as compared to normal water or distilled water.
- The last factor we need to talk about is humidity.
- This is again understood that the very fact that electrochemical corrosion is happening, we know that we need a conducting medium.
- So what happens when the it is humid, the atmosphere is humid, then there is lot of moisture which acts as the conducting medium and it is a and we can get a nice galvanic cell over there and therefore this will facilitate corrosion of the metal.
- In arid areas we find therefore that corrosion will not happen to that great extent but whenever the atmosphere.
- Is moisture laden, then we find that corrosion will be fast because it is providing the conducting medium required for the migration of ions.
- And in the next class, now we are going to start talking about corrosion control.

Example or application from the lecture

- And there what we can one very good example would be that in saline water corrosion happens at a very fast rate.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000198 · EV0000199 · EV0000200 · +1 source units in pack.json

12 Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II

Study focus: Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000202 · EV0000203

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We continue with module 6 on corrosion chemistry, and today's class we are going to talk about corrosion control and we will take up the next type of coating that is inorganic coating, wherein we will talk about anodysing and phosphating, and then we will look at what is organic coating.

Key details

- one of one metal was taken, the metal which has to be protected, and it was covered with another metal.
- Now this method is also referred to as chemical conversion coating.
- The surface of the base metal is converted into a protective coating through appropriate chemical modification.
- We have this metal, we can look at the picture here.
- Is now forming a protective film over the surface of the metal and that creates a barrier between the metal and the environment.
- External another metal which is being coated onto the surface.
- But what is the what are we doing here in inorganic coating?
- It is like we are the metal itself is undergoing corrosion and creating this protective film over the surface.
- Protective film which is formed here is an inherent part of the metal, it's an integral part of the metal itself.
- surface into an inorganic compound and hence the name inorganic coating.
- The definition would be we are going to induce oxidation by providing an oxidizing environment, and we are going to make the article into anode and then pass current.
- We know that it is the anode which undergoes corrosion and cathode remains unaffected.
- But we are making the metal into cat anode and we are passing current also.
- Will undergo oxidation, but there's a purpose here.
- The oxide which is getting formed, the metal oxide which is getting formed on the surface is going to cover the metal surface and act as a protective layer.
- can be anodized and they are told that they are usually the non-ferrous metals like aluminium, chromium, titanium.
- So we understand why aluminium particles can be anodized because aluminium when it undergoes corrosion it becomes.
- Passive and further corrosion is prevented because of this protective layer which has got formed over the surface.
- So on the surface we have Al2O3, which is going to form a nice protective film over the surface, and therefore will create a barrier and hence further corrosion is prevented.
- What are we doing here as the definition goes, we have taken the article as the anode and of course we are using a power supply here, there has to be a power supply.
- And this article is going to be connected like this.
- And here we have to when we have an anode, of course, we have to have a cathode.
- That will be some inert metal or some conducting metal has to be used.
- And this is usually aluminium or lead that is used as a cathode.

Example or application from the lecture

- But there's a small difference, and let us understand what that is.

- The difference from metallic coating is when we talked about metallic coating, we have a metal M1, and we coated it with another metal, say M2, and in this case we know that M1 is the base metal, M2 is an example.
- Let us study what is anodizing.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000202 · EV0000203 · EV0000204 · +1 source units in pack.json

Core explanation · 05:00–09:59

And we are providing the atmosphere which is oxidizing in nature.

Key details

- We are saying that we are going to induce oxidation, yes, by passing the by passing current.
- and then it will it will be done in an oxidizing environment and only such metals which can form a protective film obviously can be anodized.
- Because we know, and it's a big industry where anodizing of aluminium is done, and that passive layer is created onto the surface of aluminium by inducing the oxidation.
- So that means.
- Why are we doing it?
- We know that aluminum, anyway, when it's exposed to the when it's exposed to the environment, it is going to undergo it's going to become passive.
- Because we can control the thickness of the layer by.
- Inducing the oxidation ourselves.
- Of course, any coating when you do, we have to clean the article.
- That's exactly what is done.
- And the first step, the article is clean, decreased, and polish, and then it is taken as the anode.
- So the first step is you have chosen your anode that is the article which has to be coated.
- Sulfuric acid, phosphoric acid, oxanic acid, or it could be a mixture, and this is maintained at around 40 degrees.
- And the voltage is about 40 volts which is applied here, and what happens because of this now when you pass current, there will be an oxide layer getting formed over the surface.
- Of aluminium is oxidized to Al2O3.
- Aluminium oxide now formed on the surface is slightly porous in nature.
- The article is dipped in boiling water or steam.
- It gets converted to Al2O3HOT, but hydrated.
- And since this is a larger molecule, it is going to occupy a higher volume and all the pores are going to get sealed.
- You can see in this nice picture over here, this is the substrate.
- And aluminium oxide has got formed over the surface.
- What is done for that?
- If we dip this anodized article into hot water, boiling water, or steam, it is found that.
- It forms ALOOH or it gets hydrated, and you can see that the volume of this hydrated ALOOH is much larger, and therefore it is going to seal all the pores.

- The process of anonymizing is done for any aluminium article.
- That is also possible in the case of anodised articles.
- Because we know again that the layers are poor.
- After anodising, we are going to use the dip the article in.

Example or application from the lecture

- Let's understand the full process here.
- Let us see.

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EV0000206 · EV0000207 · EV0000208 · +1 source units in pack.json

Core explanation · 10:00–14:59

An organic dye that color seeps in through the pores and then it is sealed, so you get a nice colored finish.

Key details

- You can see here that first the surface is being cleaned and it has got anodized but there are these pores you can see the pores here now when you dip it in any paint any organic dye this color is going to get in and you will get a very good colored finish for this anodized article.
- Or another method is by depositing a metal like nickel cobalt.
- that nickel cobalt salts are very colourful, so we can deposit these metals, nickel, cobalt, etc., into the pores, and again this will give us a nice color finish.
- Such anodizing or anodized aluminium.
- We know that it is used to as an attractive, highly durable, and corrosion resistant material in exteriors for roofs, walls, building buildings, and also in window frames, office partitions, stiffen carriers, mostly.
- you will find that anodized articles are used for hookware.
- They are very popular in cookware.
- The aluminum articles are anodized and they are used for hookware.
- Which is phosphate.
- Again, this is also a chemical conversion coating or an inorganic coating matter.
- Converting the surface metal atoms into their phosphates by either by chemical or electrochemical reactions.
- like iron or steel, but now it has been extended to zinc, aluminum, and alloy alloys.
- Nevertheless for iron and steel, this is very commonly done getting a phosphate layer over the surface.
- If you want to do chemical phosphating, then it is just done by dipping the particle into a phosphate bar.
- But if it is electrolychical phosphate, then the process of electrolysis is used and current is used to get the phosphate layer onto the surface.
- So we will see study the simple method of chemical phosphating.
- What is the process?

- So what would be the components of the phosphoric so first and foremost of course we need phosphoric acid because we want the iron to react with the to form the phosphate layer.
- Now next we require another component that is the metal phosphates.
- What is their role?
- The zinc and manganese phosphates are.
- Going to increase the adherents of this coating which is getting formed.
- The phosphate layer, which is getting formed, at the same time, it is going to decrease the porosity of the coating because we are basically trying to create a film on the surface so it does film appears and also there is it is less porous so metal phosphates of zinc and manganese are used for this purpose and of course here we don't we are not using any power.
- The reaction to happen at a fast rate.
- What exactly is happening when you dip the article in this phosphing part?
- It's very clear the reaction will include what?
- The dissolution of the metal, and then there is a formation of phosphate layer which is due to the reaction with the bark solution and this phosphate layer is going to get deposited onto the surface.

Example or application from the lecture

- Now let's talk of the next method.

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EV0000210 · EV0000211 · EV0000212 · +1 source units in pack.json

Core explanation · 15:00–19:59

Phosphoric acid is going to give you Fe3 BO4 twice, which is going to get deposit on the surface and 3 hydrogen.

Key details

- In fact, it is known for its porosity.
- Any item compound usually will be porous.
- You will find that it is porous in nature.
- Very high corrosion resistance, but it is used as a base for painting.
- The article is first subjected to phosphating.
- And now when we paint on this porous layer the paint the paint will get interlocked onto the surface very strongly.
- Though it is forming a porous layer it.
- It is because this porous layer, once on this porous layer paint is applied, then it is going to give it a very good addition quality and because it is going to interlock the paint due to the pores.
- So it will be about 3 micrometers and then because it is porous, as I mentioned just now, it is going to interlock the paint very well.
- As you can understand, it has a very big industry phosphating because it is the pre-treatment which is done for all automobile bodies.
- refrigerators, washing machines, all steel sheet fabrications.

- We know this is a very common method of corrosion control.
- Most widely used as anti-corrosion coatings, and it's understood why because they keep out the air and the.
- Moisture from coming in contact with the metal surface, they're basically creating a barrier between the metal surface and the corrosion environment.
- by which we created a protective coating on the surface.
- The last class we talked about metal coating and then we talked about protect inorganic coating and then the organic coating.
- In the next class, we are going to talk about another method of corrosion control and that is about corrosion inhibitors.

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EV0000214 · EV0000215 · EV0000216 · +1 source units in pack.json

13 Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I

Study focus: Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000218 · EV0000219

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module 6 on corrosion chemistry, and today's class we are going to talk about corrosion control.

Key details

- In that we will look at metallic coating and specifically galvanization and tinning.
- When we talk of corrosion control, the first thing which comes to our mind is to create a barrier between the surface and the environment.
- So protective coating is a very good method by which we can control corrosion to a great extent.
- In protective coating, we have metallic coating, inorganic coating, and organic coating.
- Then another method of corrosion control is to use corrosion inhibitors.
- Again, we have two classes here: anodic inhibitors and catodic inhibitors.
- The other method would be cathodic protection, and in that we are going to study about two methods in there, that is sacrificial anode method and the impressed cathodic current method.
- Finally, we will study about anodic protection, which is also known as impressed anodic current method.

- The first type of coating we are going to look at is metallic coating.
- Corrosion of the base metal can be prevented by coating it with a layer of another metal over it.
- Therefore, we will have two types of coatings which are possible.
- One is called anodic metal coating where we are going to coat a layer of metal which is anodic with respect to the base metal.
- Which is cathodic with respect to the base metal.
- As I said just now, the coating metal is more active with respect to the base metal.
- Is not completely covered, it is not going to undergo pitting corrosion.
- This is a very big advantage of anodic metal coating.
- With another metal, say this is M2.
- To the base metal.
- Therefore, even if there is a gap here, we will find that if the coating is not completely covering the surface, there is a gap on the surface.
- You have studied that.
- This will not give rise to pitting corrosion because as long as the anode is in contact with the.
- Cathode, it is the anode which undergoes corrosion.
- That is the biggest advantage of having anodic metal coating.
- When we cover the metal with zinc on the surface, like zinc is acting as the coating, then this process is known as galvanization.
- It's usually on iron.
- Coating a layer of zinc on iron would be referred to as galvanization.
- What would be the various steps involved in this process?
- Make sure that the surface is absolutely clean because we know any coating when it is made on a surface, the smoother the surface, the better the coating is going to adhere to the surface.
- So the first and foremost treatment before.
- coating any surface is to make it absolutely clean.
- So in that the first step would be.

Example or application from the lecture

- Let us start with protective coatings.
- A good example would be if you have an iron article and you are covering it with zinc metal, this would be anodic metal coating and then there is another method where we which we call it as the cathodic metal coating where we are going to coat the metal with another metal.
- And an example for that would be an iron article which is coated with tin.

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EV0000218 · EV0000219 · EV0000220 · +1 source units in pack.json

Core explanation · 05:00–09:59

The iron sheet is passed to an organic solvent or a caustic solution to remove any oil or grease which may be present on the surface.

Key details

- Next, it has to be washed with dilute sulfuric acid.
- This process is known as pickling.
- The iron sheet is passed through dilute sulfuric acid so that.
- Any rust, steel, scale, or dust can be removed from the surface.
- And now the next treatment is again very important.
- It is going to pass through an aqueous solution of zinc chloride and ammonium chloride.
- What happens when we use this flux onto the surface is that it will prevent the oxidation.
- Once that is done, it is dried.
- And finally, we come to the main tank, that is the galvanizing tank, where we are going to dip the article into a zinc into molten zinc.
- Of course, if you want a coating of zinc onto the surface, it should be molten zinc, and that melting point of zinc is around 425, 430.
- And finally, excess zinc which is present on the iron sheet has to be removed by passing it through hot rollers.
- So this would be the surfa the method of galvanization, the whole process of galvanization.
- So as I told you, first we need to prepare the surface before it is galvanized.
- in with dilute estrus support again there is rinsing and then finally the flux and then it is dried and you have then it gets into the zinc bar.
- Now, this is called dry method.
- There's also a wet method where the flux need not be separate.
- What is done is the zinc bath itself is covered with this flux.
- So it's a wet method.
- That is also done in some cases.
- Galvanized iron,, we have heard of GI pipes and GI sheets.
- They are called galvanized iron sheets and pipes, which are which is used in building construction material.
- So, as I mentioned just in the beginning of the class, the advantage of this galvanization is that even if the zinc coat peels off or there are gaps on the surface of the gaps in the coating we find that the base metal iron will never get corroded because as long as zinc is present in contact with the iron it is zinc which gets corroded and not the iron article.
- Which is underneath.
- But there is also a serious disadvantage.
- What is the disadvantage?
- That galvanized articles are not used for preparing and storing food because the zinc which is present on the surface is going to dissolve in dilute acids, and that is going to produce.
- So therefore we find that the galvanized article the galvanized sheets cannot be used for storing food.
- making those storing food storing containers.
- So otherwise galvanization is very good because as long as zinc is there, we don't have any problem.
- So that means we need to look for another type of coating, and that is where we talk about the.
- The coating metal is less active with respect to the base metal.

- Means it is more noble as compared to the base metal.
- Now because it is more noble, as long as it covers the entire surface, corrosion will be much less.

Example or application from the lecture

- Let us see this picture and understand.

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EV0000222 · EV0000223 · EV0000224 · +1 source units in pack.json

Core explanation · 10:00–14:59

But there is a serious problem here, which we have talked about many times.

Key details

- If the coating is not uniform.
- Metal, if there are gaps then now what has happened because the coating is cathodic with respect to the base metal, this gap is going to expose the metal M1, which is now acting as the anode.
- So you have a case of small anode and large cathode, just like we understood in during pitting correct studying pitting corrosion, when we were studying pitting corrosion, and therefore this is going to cause severe fitting corrosion and the metal actually underneath is going to get more damaged than if it was left uncovered.
- How would we define tinning?
- Coating a layer of tin over iron will be called the tinning process.
- Again we understand the article.
- Has to be dipped.
- Now this method is known as the hot dipping method.
- In fact, even the galvanization which we did just now is known as the hot dipping method.
- It's understood why because we are dipping the article in the molten zinc which is hot.
- So therefore we need to dip the article in molten tin, but before that again we need to.
- Take care, I mean we need to do the surface preparation, and here specifically, because we know if there are gaps left on the surface, it is going to result in intense corrosion or pitting corrosion of the article which we want to protect.
- Therefore, we need to take special precautions.
- First the steps are same.
- We are going to pass it through organic solvent or caustic solution to remove the oil or grease present on the surface, and then it is washed with dilute H2O4.
- This process is called pickling to remove any rust, scale, dust, etc.
- And then next step is very important where it is going to be treated with zinc chloride flux.
- It is going to make sure that the tin coating adheres to the surface.
- And then it is dipped in the in molten tin.
- Centigrade, so that it's in the molten form, and then your article is dipped in it.
- Obviously, it is going to get a tin coating.
- Excess tin now is removed.

- By passing through hot rollers, and now it is then immersed in palm oil.
- It prevents the surface oxidation of the freshly coated tin.
- Here we have not shown the organic solvent or the caustic solution treatment.
- This is where the pickling is taking place.
- After pickling, you can see that the sheet.
- It is passing through a bath of zinc chloride flux and then it comes to the molten tin area, and finally it is going to the palm oil, which is going to cover the surface and prevent oxidation of the freshly coated tin.
- Then it goes to the rollers and.
- All the excess tennis report and this is the process of tinning.
- What are the applications?
- It's understood that tin coated steel.
- Why did we talk about this process in the first place?
- Because we know galvanized sheets cannot be used for making containers which store food.
- Make fruits, pickles, etc.
- We know that because it's highly they will be slightly acidic in nature, and they could.

Example or application from the lecture

- So an example for this cathodic metal coating would be coating of tin or nickel metal over iron.
- Let us look at this picture and understand.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000226 · EV0000227 · EV0000228 · +1 source units in pack.json

Core explanation · 15:00–19:59

That would provide the right environment for corrosion.

Key details

- And also, if it was zinc, it would have dissolved in this, but here it may we have used tin coating, so it is safe.
- And obviously, that would be the biggest advantage of tinning over galvanization.
- Tin coated steel can be used for making containers to store food, but of course, we know there is a serious disadvantage that there will be intense and accelerated corrosion if.
- Even a small crack is formed, or while coating any gaps left on the surface, this would result in pitting corrosion.
- due to the formation of small anode and large cathode area.
- So in the next class we are going to talk about.
- Another method of protective coating and that is inorganic coating.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000230 · EV0000231 · EV0000232 · +1 source units in pack.json

14 Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II

Study focus: Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000234 · EV0000235

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module six on corrosion chemistry and today's class we are going to talk about yet another method of corrosion control and that is cathodic protection.

Key details

- In that there are two ways, one is known as the sacrificial anon method, the other is known as impressed cathodic current method.
- To be protected into cathode.
- This is coming from the electrochemical theory of corrosion, where we understand that it's only the anode which undergoes corrosion, whereas the cathode remains unaffected.
- The structure to be protected is completely converted into cathode and no part of it is allowed to act as anode.
- And we know that since cathode does not undergo corrosion, so the structure is completely protected from corrosion.
- One is known as the sacrificial annode method, and the other is the impressed current method.
- Now in this method.
- Remember that we have to convert the metal into cathode.
- One good way would be if it is connected to a more active metal.
- In contact with the cathodic metal, the cathodic metal gets protected completely as long as the anodic metal is present in contact with the cathodic metal, the it is the anode which undergoes corrosion.
- Concept is used here.
- They are zinc and magnesium.
- most of the times and these are referred to as auxiliary anode.
- Now why do we call it as the sacrificial anode method?
- This is referred to as the sacrificial anode method for the simple reason that in order to protect the structure, metallic structure, we are sacrificing the metal which is acting as the anode.
- Hence, this is known as the sacrificial anode method.
- The only problem with this method is that as the anodes get exhausted, they have to be replaced periodically because we have to understand that the cathode structure now is

protected only.
- as long as it is in contact with an anode metal.
- Magnesium and zinc bars which are fixed to sides of ocean going ships.
- Ship is getting protected just by attaching a magnesium or zinc bar onto the surface, of course.
- It is not that we put a small zinc bar and a whole big ship is protected.
- Yes, it is done by calculation and at regular intervals these magnesium you can see here also so many of them are placed here.
- Regular intervals these magnesium or zinc bars are pr are attached to the body of the ship and that is going to protect the big huge ship from corrosion.
- very pipeline, we find that the pipeline is protected from corrosion.
- Again, it has to be kept at regular intervals.

Example or application from the lecture

- Let us understand what we do in sacrificial annode method.
- Suppose we have.
- So this would be a very good example for sacrificial anode method.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000234 · EV0000235 · EV0000236 · +1 source units in pack.json

Core explanation · 05:00–09:59

a pipeline which has to be protected and therefore we ensure that it is always in contact with the zinc wire so that the zinc undergoes corrosion and the pipeline metal is protected from corrosion.

Key details

- Usually it will be steel.
- What are the advantages of this method?
- It's a very simple method and the insulation cost is very low and it does not require any power supply.
- The magnesium bar or the zinc bar has to be just in contact with the structure to be protected.
- would be the disadvantage.
- It involves recurring expenditure because we need to replace the consumed anode.
- So it's a very simple method by which we can protect the structure from corrosion and we are not just talking about protecting it, it is completely unaffected by corrosion, it's only the anode which is undergoing corrosion here and the structure to be protected is absolutely unaffected by this by corrosion.
- Here we are going to use direct current.
- Now again another way of making the protected structure into cathode would be if we can connect the external connect an external store source the cathode terminal of the external source to the protected structure so if you are forcing it to behave like the cathode because it's connected to the cathode terminal of the external source and the anode will be connected to another metal over it.

- We have to provide current of course but this now has a.
- The cathode terminal will be connected to the cathode to the protected structure so that you when you pass current you are forcing your metal to behave as cathode and that is what we talk about in cathodic protection.
- Of course you have to remember if.
- The anodes are not inert, in that case we need to keep replacing them.
- But if we use a most common use is that of the graphite where it is.
- It does not it is an inert anode.
- therefore now your protected metal, the metal which you want to protect that is acting as the cathode, so it does not undergo corrosion and also your anode being inert remains unaffected.
- Here we have to have a backfill of coke or bentonite which is used to improve the efficiency of the anode.
- Place the have a backfill around it.
- as I said to our stationary structures like bridges etc this is very commonly done then offshore structures where we have these oil wigs etc there also this protection is used.

Example or application from the lecture

- Let us look at the next method that is the impressed current method.
- First let us concentrate on the cathode.
- Now let us see an example here.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000238 · EV0000239 · EV0000240 · +1 source units in pack.json

Core explanation · 10:00–14:59

And around this anode we have the backfill which is usually coke or coke and bentonite.

Key details

- It's a type of clay.
- That one installation can protect a large area of the metal, whereas in sacrificial anode we said that it had the these anode have to be placed at regular intervals.
- But here one installation can protect a large area of the metal.
- Of course, one is that it is rather expensive because we need a very high current for the safe protection of the structure.
- That is the first thing we understand.
- If the impressed current is not uniform on the entire surface, then.
- What will happen?
- There will be local anode and cathode form because the current is not uniform over the entire surface.
- There will be local anode and cathode and that will give rise to localized corrosion of the protected metal.
- If hydrogen liberation is the cathodic reaction.

- Yes, that does happen, but we know that though the metal itself is not involved in the reaction, the environment has to take part in the.
- H plus taking up electron and giving you half H two.
- If this is the main reaction, or even in a neutral atmosphere, it is possible that hydrogen liberation can take place.
- We have studied the reaction.
- Hydrogen evolution if it is taking place either in an acidic environment like the reaction I've written here or it could also happen in the neutral environment if that happens then a lot of hydrogen is getting produced and that will result in hydrogen embrittlement.
- This being very small, hydrogen molecule is a very small molecule, so it enters the interstitial space of the metal lattice, and when the pressure starts building up, more and more hydrogen is getting involved.
- And hydrogen liberation.
- Whether it is an acidic medium or it is in neutral or alkaline medium, we know that hydrogen liberation could be an important reaction which will take place at the cathode, and in that case, what is going to happen?
- It will result in hydrogen.
- So these are some serious limitations of the cat cathodic protection and especially when we are using the impressed anode cathodic current method.
- Of course this hydrogen embrittlement can take place even when we are using any cathodic protection like even in the sacrificial anode method but yes there is a possibility that hydrogen embrittlement will take place when.

Example or application from the lecture

- Let us have a little understanding of this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000242 · EV0000243 · EV0000244 · +1 source units in pack.json

Core explanation · 15:00–19:59

Whenever we use cathodic protection.

Key details

- These this is about the how we are going to protect the metal structure using cathodic protection.
- The entire structure has been made into cathode and we have an anode which is undergoing corrosion if it is active, if it is an inert and anode then it is also unaffected by corrosion.
- So now in the next class we are talking going to talk about another method of protection and that is anodic protection.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000246 · EV0000247 · EV0000248 · +1 source units in pack.json

15 Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I

Study focus: Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000250 · EV0000251

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module six on corrosion chemistry and in today's class we are going to study about corrosion control.

Key details

- We will look at another method of corrosion control and that is using corrosion inhibitors.
- There are two types of corrosion inhibitors, anodic corrosion inhibitors and cathodic corrosion inhibitors.
- They are chemical substances which are added in small quantities to the corrosive environment in order to decrease the rate of corrosion.
- Environment and based on that, and also the nature of the metal, which is which needs to be protected, we are going to choose a corrosion inhibitor and use it, add it to the corrosive environment, and this should be added only in small quantities.
- Now because we know there are two different reactions which happen at dianode and cathode, so there would be different types of corrosion inhibitors.
- Basically, they are going to inhibit either the anodic reaction or they'll inhibit the cathodic reaction and therefore they will help us to control corrosion.
- But the only.
- Problem with this corrosion inhibitor is that it can be used only in a confined environment.
- That can be understood because the concept is we are adding some substance into the corrosive environment.
- It has to be an enclosed environment so we say it is used only in confined environment.
- Action of the corrosion inhibitor obviously will depend on the article which has to be protected as well as the corrosive environment.
- We need to study the specific case.
- what is the nature of the metal, what is the corrosive environment, and then decide what corrosion inhibitor would be suitable to protect that article from corrosion.
- We know metal is getting oxidized forming mn plus n electron n electrons.
- If we achieve that, then we are going to retard the corrosion process.
- This is done with the help of addition of large anions.
- It requires large anions because large anions like chromatic usually will form precipitate with these metal ions.

- What is the reason for that?
- We know that the metal surface is in the is acting as the anode.
- They are going to form a precipitate as I said just now, and they are going to cover the surface of the anode.
- Now, if this covering is not uniform, if there are gaps, then your anode is getting exposed, and therefore this will.

Example or application from the lecture

- Let us understand what exactly are corrosion inhibitors.
- Let us understand what is an anodic inhibitor.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000250 · EV0000251 · EV0000252 · +1 source units in pack.json

Core explanation · 05:00–09:59

small amount of the anode area is getting exposed because the precipitate is not completely covering the.

Key details

- Surface and therefore this will lead to intense corrosion.
- So, therefore, whenever anodic inhibitors are used, we have to make sure that we add sufficient amount of the inhibitor.
- Inhibitor we are using here so sodium chromate sodium tungstate.
- Now when we come to the cathode, we know from the electrochemical theory of corrosion that there can be different types of reactions which will take place because the metal itself is not undergoing reduction.
- It cannot undergo reduction.
- And what were the two types of reactions?
- we had classified into they were cathodic reactions were either it could be liberation of hydrogen or it could be absorption of high oxygen.
- So that means if we want to control corrosion, then we need to look at ways by which either we can stop the liberation of hydrogen or we can stop the absorption of oxygen.
- Now because the cathodic reaction has become slow, it is going to in turn slow down the anodic reaction.
- And once anodic reaction is not taking place, we know that the corrosion is under control or you have inhibit corrosion to a good extent.
- How do we inhibit the liberation of hydrogen?
- It can be done in two different ways.
- This reaction is happening at the cathode and one of the reasons for the reaction would be that H plus ions are being reduced at the cathode.
- Organic compounds containing nitrogen and sulfur.
- How are they doing it?
- All these are widely used as corrosion inhibitors and what is their role?
- They are going to prevent the diffusion of H plus ions to the cathode.

- They get absorbed on the cathodic region they get absorbed on the cathodic surface and form a protective film.
- Another way of doing it is also liberation of hydrogen can be prevented by increasing the hydrogen overvoltage on the metal surface.
- Now when we studied factors affecting corrosion rate, one important factor which we studied was hydrogen over voltage and what did we understand there that if the hydrogen over voltage is high then the liberation of hydrogen is going to be less and therefore we can control we can reduce the rate of reaction or the rate of.

Example or application from the lecture

- First, let us take up hydrogen liberation.
- Let us study the first one where we are going to prevent the diffusion of H plus ions to the cathode.
- First let us look at some examples.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000254 · EV0000255 · EV0000256 · +1 source units in pack.json

Core explanation · 10:00–14:59

Corrosion is going to be less.

Key details

- What we have to do is we have to increase the hydrogen over voltage over the cathodic surface.
- It is done with the help of oxides of arsenic and antimony.
- What happens is they are going to deposit arsenic and antimony onto the cathodic region.
- Arsenic and antimony have very high.
- Hydrogen over voltage.
- Now since you have raised the hydrogen over voltage of that particular surface, the liberation of hydrogen becomes difficult and therefore corrosion is going to the rate of corrosion is going to fall down.
- How do we inhibit oxygen absorption?
- And that should not be very difficult.
- All you have to do is oxygen has to be removed.
- It is going to react with the oxygen and it will form nitrogen and H2O.
- You would have removed the oxygen from the atmosphere.
- What's a scavenger?
- It is going to eat up the oxygen.
- So oxygen scavenger is being used here, it's going to take away all the oxygen from the atmosphere, and therefore, when there is no oxygen, obviously oxygen absorption will not.
- happen meaning oxygen is not available to undergo the reduction reaction and hence you would have brought down the extent of cathodic reaction.
- and therefore we can understand that there is a loss of electrons and therefore now we know that the reducing agents when we add it is going to take up the oxygen and that will help us

to remove oxygen from the atmosphere and therefore it will reduce the cathodic reaction.
- Another way of reducing or removing the oxygen is or reducing oxygen absorption is by decreasing the diffusion rate of oxygen to the cathode.
- We are using large cations like zinc sulfate, zinc and magnesium.
- So the salts used are zinc sulphate and magnesium sulphate nickel sulphate etc why these salt because their cations are large zinc and mg2 plus zinc 2 plus mg2 plus now these are going to migrate towards the cathodic surface and react with the hydroxyl ions.
- Again this is a precipitate.
- Magnesium also reacts with hydroxide and it forms zinc.
- Magnesium hydroxide which is a precipitate.
- Now these hydroxides again they are getting deposited over on the cathodic sites and when that happens a protective film is formed over the cathodic sites and therefore oxygen cannot come in contact with the cathode and take up electrons and this is how you are preventing the absorption of oxygen.
- The anodic inhibitors and the cathodic inhibitors, how they act.

Example or application from the lecture

- Arsenic and antimony oxide, say for example, these will be the corrosion inhibitors.
- Let us talk of oxygen absorption.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000258 · EV0000259 · EV0000260 · +1 source units in pack.json

Core explanation · 15:00–19:59

In the case of cathodic inhibitors, depending on which reaction we are going to stop, if it is liberation of hydrogen, then we use urea, thiourea.

Key details

- And another was to increase the hydrogen over voltage that was also done with the help of arsenic oxide and antimony oxide.
- And there was another set of cathodic inhibitors where you could use oxygen scavengers like hydrazine or a reducing agent like sodium sulfide and also if you want to prevent oxygen from coming in contact with the cathode surface we could use zinc sulfate, magnesium sulphate etc.
- Now this is important, they will contaminate the environment because we know that corrosion inhibitors are substances we are adding directly into the environment.
- Now these are toxic and therefore they cannot be used in systems which come in contact with humans.
- And the last fall, which we have already understood, is that they can be used only in closed system.
- We cannot use them in open system, but only in closed system in which the corrosion corrosive environment is either.
- Contained or it is recirculated.

- Nevertheless, we have to study the nature of the metal, the and corrosive environment, and then decide what type of corrosion inhibitor we are going to use for that particular system.
- So that is about corrosion inhibitors and in the next class we are going to talk about another method of corrosion control and that is cathodic protection.

Example or application from the lecture

- And many of these inhibitors are toxic, like for example arsenic oxide.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000262 · EV0000263 · EV0000264 · +1 source units in pack.json

16 Source slide deck: Nernst equation and its derivation

Study focus: Source slide deck: Nernst equation and its derivation. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000299 · EV0000300

SHORT-NOTE RECAP

Study notes · Source slide deck: Nernst equation and its derivation

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING CHEMISTRY (UE26CY151 A,B)

- Department of Science and Humanities

Slide 2 · ENGINEERING CHEMISTRY

- Nernst equation & it's derivation

Slide 3 · ENGINEERING CHEMISTRY Electrochemical equilibria

- Nernst equation for a single electrode
- Energy available per unit charge, i.e., electrode potential because electrode potential = energy/unit charge = E
- Therefore , Wmax = nFE ; Δ G = - nFE

Slide 4 · ENGINEERING CHEMISTRY

- A thermodynamic equation which relates reaction quotient and decrease in free
- Δ G = ΔGo + RTlnQ , where Q is the reaction quotient The reaction quotient for the reaction is, Q= [M]/[Mn+]
- Where , Eo = Standard electrode potential, n = number of electrons exchanged in the redox reaction, R = Gas constant. 8.314 JK-1 mol-1 , T = temp in Kelvin,

Slide 5 · ENGINEERING CHEMISTRY Electrochemical equilibria

- since [M] = 1 for pure substances,
- Nernst equation may also be used to calculate. emf of electrochemical cells. For the cell reaction
- n= no. of e -s transferred, Eocell = std. emf of the cell

Slide 6 · Department of Science and Humanities

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000299 · EV0000300 · EV0000301 · +15 source units in pack.json

17 Source slide deck: Experiential learning Based on corrosion chemistry

Study focus: Source slide deck: Experiential learning Based on corrosion chemistry. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000686 · EV0000687

SHORT-NOTE RECAP

Study notes · Source slide deck: Experiential learning Based on corrosion chemistry

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING CHEMISTRY (UE26CY151 A,B)

- Department of Science and Humanities

Slide 2 · ENGINEERING CHEMISTRY

- The rate of a chemical reaction, in general, increases with rise in temperature
- Increase in conductance of the medium- ions migrate fast at higher temperature.
- However, if corrosion is due to dissolved gases like O2, CO2, etc., the rate of corrosion decreases with increase in temperature as the solubility of these gases in the medium decreases

Slide 3 · ENGINEERING CHEMISTRY

- In general, lower the pH of the corrosion medium, higher is the
- At low pH ,concentration of H+ ions is high; H+ ions help for cathodic reaction facilitating anodic oxidation of the metal
- However, some metals like Al undergo fast corrosion in

Slide 4 · ENGINEERING CHEMISTRY

- As the conductivity of the corrosion medium increases, the corrosion rate also increases
- Higher the conductivity of the medium, faster the ions can migrate between the anodic and cathodic regions of the corrosion cell
- Faster will be the exchange of electrons at the electrode surfaces; corrosion rate increases

Slide 5 · ENGINEERING CHEMISTRY

- Experiential Learning – UNIT 2 (supplementary material)
- Corrosion rate increases with increase in the humidity in the medium.
- The presence of moisture in the atmosphere provides the conducting medium for the formation of galvanic cell, facilitating corrosion of the metal.

Slide 6 · [Image text, OCR]

- Department of Science and Humanities

EV0000686 · EV0000687 · EV0000688 · +15 source units in pack.json

18 Historical question-source pages · unit mapping candidates

Study focus: Historical question-source pages · unit mapping candidates. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000704 · EV0000705

SHORT-NOTE RECAP

Study notes · Historical question-source pages · unit mapping candidates

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Archive page 3 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.5; source text/OCR may be partial. (4.0 Marks) (5.0 Marks) (4.0 Marks) (2.0 Marks) 2.b. For a Pb-Ag system 1. Why is condensed phase rule applicable to this system? 2. Explain how silver can be extracted from Argentiferrous lead? 3.What is the Eutectic composition and Eutectic temperature? 2.c. A silver electrode containing 0.15M silver nitrate and copper electrode containing 0.19M copper sulphate were coupled using salt bridge. Write the cell representation, half cell reactions, calculate E , E at 25 C (Give E = 0.8V, E = 0.34V) 2.d. A Satuarated calomel electrode and glass electrode assembly was used to determine unknown pH. In the above cell following potentials are recorded a. 0.2836V in contact with buffer solution of pH = 4. b. 0.1949V in contact with solution of unknown pH. Write cell representation and calculate unknown pH (Given Esce = 0.24V) 2.e. i. Mention the factors affecting the conductance of a solution. 1M ii. In conductometric titration experiment, 25mL of acid mixture is titrated against 1M sodium hydroxide (NaOH) and the graph reveles V1 and V2 are 3.2 mL and 7.3 mL of NaOH respectively, Calculate the normality of acetic acid. 1M ocell cell o oAg /Ag + oCu /Cu 2+

Archive page 5 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.4167; source text/OCR may be partial. (2.0 Marks) (5.0 Marks) (4.0 Marks) 3.e. For a weak acid, pH = pKa, Substantiate using Henderson-Hasselbalch equation. 4.a. i. "Electro chemical Theory has provided satisfactory explaination for corrosion of metals" Justify the statement by taking iron as a demonstration sample metal, give the reactions of anode and cathode at different corroding environments. -4M How does iron corrods in presence of excess of oxygen and limited sypply of oxygen? - 1M 4.b. i. Identify the type of corrosion of the metal pipe in the picture ii. Briefly reason out the corroding environment and write electro chemical reactions at anode and cathode in the given picture - 3M

Archive page 6 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.8333; source text/OCR may be partial. (4.0 Marks) (4.0 Marks) (3.0 Marks) (4.0 Marks) 4.c. i. What are corrosion ihhibitors? Describe in a sentence or two how they control the corrosion rate - 2M ii. Anodizing and Phosphating are two techniques to protect a metal from corroding by coating the metal with an inorganic coating. Write the two compounds and the relevant chemical reactions for the anodization of aluminium and phosphating of iron. - 2M 4.d. i. Galvanization and tinning are two popular methods of protecting a metal by coating with another metal. If the reduction potentials of iron, Zinc and tin are -0.44V, -0.76V and -0.14V respectively, then comment on the situation;

corrosion of iron would be observed in case a section of the protecting layer wears off. Briefly discuss the reason. ii. Write the corrosion cell that represents caustic embrittlement. What could go wrong with a boiler if caustic embrittlement is not taken care of? 4.e. i. In naphthalene–biphenyl system mention the number of components and phases at eutectic point. -2M

ii. Comment on the following statement with respect to viscosity: "In an Ostwald viscometer, two liquids, A and B, take 60 seconds and 90 seconds respectively to flow between two marks of two identical capillaries" 5.a. What are biodegradable polymers? Give the two step process of degradation of polymers. Mention the factors influencing the degradation of polymers

Archive page 4 from 11__1zfeA5C-J7.pdf; unit match candidate score 0.5833; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 3.d. What happens when ammonia solution is added to cupric ions during 'Colorimetric estimation of copper in copper sulphate' experiment. Why is the analysis done at 620nm wavelength? 4.a. (i) Under what conditions can hydrazine be used as a corrosion inhibitor? How does it bring about corrosion inhibition, give the relevant reaction. (2 marks) (ii) What would be the effect of using a depolariser like cupric ion on the rate of corrosion reaction in a system that is cathodically polarised? (2 marks) (iii) What would be the cathodic reaction as per 'Electrochemical Theory of Wet Corrosion' if the corrosive medium is neutral and devoid of oxygen. (2 marks) 4.b. (i) What kind of corrosion would take place if steel bolts and nuts are present on a huge copper sheet? How would be the rate/severity of corrosion be affected by the ratio of anodic to cathodic area in this case. (3 marks) (ii) How would addition of insufficient amount of Na CrO (sodium chromate) to a confined corrosive environment affect the rate of corrosion. (3 marks) 4.c. (i) Why should the stainless steel tank being protected by the anodic protection technique be always maintained at a voltage corresponding to passive region? What would happen if the voltage strays into the trans passive region? (3 marks) (ii) A sample of brass of area 12.5 cm when exposed to a given electrolyte gave a corrosion current of 0.25 Amperes during Tafel studies, Calculate the rate of corrosion in mpy if the density and the equivalent weight of the brass sample is 8.73 g/cm and 32.068 respectively. (3 marks) 2 4 2 3

Archive page 3 from 1__1rBrYYqw3s.pdf; unit match candidate score 0.75; source text/OCR may be partial. (5.0 Marks) (6.0 Marks) (2.0 Marks) (6.0 Marks) 2.c. i) Explain the effects of following factors on corrosion of metal a) Nature of corrosion product b) Hydrogen overvoltage ii) Identify the type of corrosion taking place, when steel pipe connected with copper plumbing. 2.d. i) Explain the process of anodizing with the help of a diagram and relevant reaction. ii) How is sealing done for anodized articles ? 2.e. (i) Mention the role of sulphuric acid in the estimation of ferrous iron by potentiometric titration method. ii) Why pH is equals to pKa at half equivalence point in the pH metric titration of weak acid vs strong base ? 3.a. i) what are reserve batteries ? Give an example. ii) Calculate the capacity (Ah), energy density (Wh/kg) and power density (W/kg) of a battery if 2.4 g of lithium is stored in the battery and the battery lasts for 120 mins. The battery weighs 115 g and gives steady voltage of 3.4 V. (Given: F = 96500 C/mol, atomic mass of lithium = 7)

Archive page 3 from 3__1MwY1CTl_j.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 3/7 (4.0 Marks) (5.0 Marks) (8.0 Marks) 2.b. A decinormal calomel electrode is used to determine the potential of the following redox electrode: Pt/Cu (0.58M),Cu (0.08M). (i) write

the cell representation (ii) write the half-cell reactions at both the electrodes (iii) calculate E and E at 298K (Given: E = 0.281 V, E = 0.153V) 2.c. (i) A glass electrode is coupled with saturated calomel electrode to measure unknown pH. The cell potentials measured are 0.215 V and 0.385 V in contact with a solution of pH7 and with solution of unknown pH respectively. Calculate the pH of unknown solution. Given: E = 0.244 V (2M) (ii) Discuss the construction of glass electrode with a labelled diagram (3M) 2.d. (i) Briefly describe the process of galvanization of iron (3M) (ii) Describe the cathodic protection by sacrificial anode method with its advantages and disadvantages (3M + 2M) 2+ + 0cell cell 0Hg/Hg2Cl2/Cl- 0Cu2+/Cu+ SCE

Archive page 2 from 4__1eKMFaJ2lx.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 2/7 (6.0 Marks) (3.0 Marks) (6.0 Marks) 1.d. (i) Draw a neat labeled phase diagram of Lead-Silver system indicating the different areas and lines. Give the temperature and composition values corresponding to eutectic point. (ii) For Iron-Carbon system, mention the phases that are equilibrium with each other, corresponding temperature and weight % of carbon at the peritectic point. 1.e. i) Mention the factors affecting the conductance of a solution. ii) In conductometric titration experiment, 25 cm acid mixture is titrated against 1N NaOH and the graph revels V and V values at 2.8 cm and 6.5 cm of NaOH respectively. Calculate the normality of hydrochloric acid and acetic acid. 2.a. For the cell Pt/Hg/Hg Cl /Cl (0.1N)//Fe (0.05M), Fe (0.5M),/Pt i) Write the anodic, cathodic and overall cell reaction. (ii) Calculate E and E at 25 C (Given: E = 0.281V, E = 0.77V) . 3 1 2 3 3 2 2 - 3+ 2+ oCell Cell o o calomel oFe3+/Fe2+

Archive page 3 from 4__1eKMFaJ2lx.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 3/7 (6.0 Marks) (5.0 Marks) (6.0 Marks) 2.b. (i) What is ion selective electrode ? Write the cell representation when a glass electrode is coupled with a calomel electrode. (ii) When a buffer of pH = 4.0 is used an emf of 0.786 V is recorded at 298 K. At the same temperature when a solution of unknown pH is used an emf of 0.825 V is recorded. Calculate the pH of the unknown solution. (Given: E = 0.2422 V) 2.c. i) Explain the effects of following factors on corrosion of metal a) Nature of corrosion product b) Hydrogen overvoltage ii) Identify the type of corrosion taking place, when steel pipe connected with copper plumbing. 2.d. i) Explain the process of anodizing with the help of a diagram and relevant reaction. ii) How is sealing done for anodized articles ? SCE

Archive page 3 from 5__1pIeTzocnU.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 3/7 (6.0 Marks) (6.0 Marks) (6.0 Marks) 2.b. i) What is the asymmetric electrode potential for a glass electrode? ii) A glass electrode dipped in a solution of pH of 4 offers an EMF of 0.2066 V with SCE at 298 K. Dipped in a solution of unknown pH, at the same temperature, in contact with SCE, recorded an EMF of 0.1076 V. Calculate the pH of the solution if ESCE is 0.2412 V 2.c. (i) Consider two situations: one where Iron (Fe) is in contact with copper (Cu) and the other where iron is in contact with silver (Ag). Given the standard reduction potentials of Fe, Cu, and Ag as -0.44 V, 0.34 V, and 0.8 V respectively, determine which scenario would lead to a faster corrosion of Fe and explain the reason. (ii) If the metal ions liberated at the anode accumulate in the anodic region, what impact would this have on the corrosion rate? Explain the observed effect. (iii) Describe the conditions under which oxides of antimony and arsenic can be employed for corrosion

control. Explain how these oxides contribute to a reduction in the corrosion rate. 2.d. 1. What is anodizing of aluminum? How can one produce anodized aluminum articles? 2. How is sealing done during anodization, Give the relevant reaction. 3. How can colored anodized articles be produced?

Archive page 2 from 6__1H6QH_PVNf.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 2/7 (6.0 Marks) (3.0 Marks) (6.0 Marks) 1.d. (i) Draw a neat labeled phase diagram of Lead-Silver system indicating the different areas and lines. Give the temperature and composition values corresponding to eutectic point. (ii) For Iron-Carbon system, mention the phases that are equilibrium with each other, corresponding temperature and weight % of carbon at the peritectic point. 1.e. i) Mention the factors affecting the conductance of a solution. ii) In conductometric titration experiment, 25 cm acid mixture is titrated against 1N NaOH and the graph revels V and V values at 2.8 cm and 6.5 cm of NaOH respectively. Calculate the normality of hydrochloric acid and acetic acid. 2.a. For the cell Pt/Hg/Hg Cl /Cl (0.1N)//Fe (0.05M), Fe (0.5M),/Pt i) Write the anodic, cathodic and overall cell reaction. (ii) Calculate E and E at 25 C (Given: E = 0.281V, E = 0.77V) . 3 1 2 3 3 2 2 - 3+ 2+ oCell Cell o o calomel oFe3+/Fe2+

Archive page 3 from 6__1H6QH_PVNf.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 3/7 (6.0 Marks) (5.0 Marks) (6.0 Marks) 2.b. (i) What is ion selective electrode ? Write the cell representation when a glass electrode is coupled with a calomel electrode. (ii) When a buffer of pH = 4.0 is used an emf of 0.786 V is recorded at 298 K. At the same temperature when a solution of unknown pH is used an emf of 0.825 V is recorded. Calculate the pH of the unknown solution. (Given: E = 0.2422 V) 2.c. i) Explain the effects of following factors on corrosion of metal a) Nature of corrosion product b) Hydrogen overvoltage ii) Identify the type of corrosion taking place, when steel pipe connected with copper plumbing. 2.d. i) Explain the process of anodizing with the help of a diagram and relevant reaction. ii) How is sealing done for anodized articles ? SCE

Archive page 4 from 7__16pEjcSTVO.pdf; unit match candidate score 1.0; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 3.d. What is Beer- Lambert's law? 4.a. What is differential aeration corrosion? Draw a suitable diagram to illustrate the same, write the corrosion reaction (both anodic and cathodic) assuming the medium to be neutral and oxygen to be present.. 4.b. How do the below mentioned factors affect the rate of corrosion- (i) Rate of anodic to cathodic area (ii) pH of the medium (iii) Nature of metal 4.c. (i) Explain sacrificial anodic protection method of corrosion control with the help of a diagram. (ii) Explain the steps involved in galvanization technique of corrosion control.

Archive page 5 from 8__1015pA2nzY.pdf; unit match candidate score 0.4167; source text/OCR may be partial. 3/25/25, 4:03 PM ESA - Dec 2024 - UE22CY151B/UE22CY151A (set- 1) about:blank 5/7 (6.0 Marks) (5.0 Marks) (4.0 Marks) 4.a. What is anodizing? Explain the process of anodizing and give its application. 4.b. i) Why do the metals undergo corrosion? ii) Two metals such as iron and tin are in contact with each other and exposed to the corrosive environment, name the type of corrosion and explain with the relevant reactions. 4.c. Discuss how the following factors affect corrosion: i) Nature of the metal ii) pH

Archive page 6 from 9__14fBQb1-hv.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 11/8/24, 1:02 PM ESA - DEC 2023 - UE22CY151B/UE22CY151A (set- 1) about:blank 6/9 (5.0 Marks) (4.0 Marks) (4.0 Marks) 4.a. i. Explain Electrochemical throry of Corrosion, with suitable example. (4M) ii. Comment on yellow rust and black rust of iron (1M) 4.b. i. "Corrosion of Iron (Fe) is faster when it is in contact with copper (Cu)" reason out the fact. (2M) ii. How does nature of corrosion product influence the rate of corroiosn? (2M) 4.c. i. what are cathodic inhibitors? (1M) ii. Mention any one chemical compound used an cathodic inhibitor which inhibit the hydrogen liberation at cathode. (1M) iii. Mention any two limitations of Cathodic inhibitors (2M)

Archive page 2 from Chem-EC[2025]__1ZtWpoG9Dv.pdf; unit match candidate score 0.75; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 2) about:blank 2/6 (7.0 Marks) (6.0 Marks) (6.0 Marks) 1.d. (i) What is computational chemistry? Give any three applications of computational chemistry. (4 marks) (ii) During the conductometric titration of a mixture of strong and weak acid (HCl and CH COOH) against a strong base (NaOH), account for the nature of the conductance vs Volume of NaOH graph obtained. (3 marks) 2.a. For the below mentioned cell- Ag/AgCl/Cl−1(0.1) //Fe2+(0.02), Fe3+(0.1)/Pt Write the anodic and cathodic reaction, E and E values. Given : The standard reduction potential values of E0 Fe 3+ /Fe 2+ =0.77V and E0 Ag/AgCl/Cl −1 = 0.28V, R=8.314 J/K/mol and F=96500 C/mol 2.b. (i) When a metal M is in contact with solution containing its ions M and if ionization is faster than deposition, explain the formation of the electrical double layer at the interface of metal and solution. (3 Marks) (ii) Derive an expression for the 'Boundary Potential' established when a glass electrode is immersed in an external solution of H ions. Assume that the concentration of H ions in the external solution is c1 and the concentration of H ions in the internal reference solution is c2. (3 Marks) 3 0cell cell n+ + + +

Archive page 3 from Chem-EC[2025]__1ZtWpoG9Dv.pdf; unit match candidate score 0.9167; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 2) about:blank 3/6 (6.0 Marks) (7.0 Marks) (6.0 Marks) 2.c. (i) If overvoltage of hydrogen gas is high on a metal surface, how does this affect the rate of corrosion of metal. (Assume that the cathodic reaction is hydrogen evolution) (2 marks) (ii) Ship sailing in the sea or docked in the yard for a long time suffers from which type of corrosion. Give the anodic and cathodic regions formed and give the corresponding cathodic reactions. (4 marks) 2.d. (i) Give examples of - (4 marks) A corrosion product that offers protection An anodic corrosion inhibitor A cathodic protection technique for preventing corrosion The flux used in galvanization (ii) In the potentiometric titration experiment for estimation of ferrous iron, name the indicator and reference electrode. Justify. (3marks) 3.a. (i) A battery using 15 g Zinc as anode and discharges a constant current for 2 hours.The voltage of the battery is 1.9 V and the total weight of battery is 50g. Calculate the energy density of the battery in Wh/Kg. (Given : molar mass of Zn = 65 g, F = 96500 C/mol and no. of electrons involved in the redox reaction is 2.) (4 marks) (ii) Give the anodic and cathodic reactions taking place in a zinc-air battery. (2 marks)

Archive page 2 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 0.5; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 2/7 (4.0 Marks) (2.0 Marks) (6.0 Marks) 1.d. Derive phase rule for a system of C components and P phases which are in equilibrium with each other. (4M) 1.e. i) Mention any two factors

affecting the conductance. (1M) ii) Write any two advantages of potentiometric titration. (1M) 2.a. Derive Nernst equation for the given cell and calculate the emf of the cell. 2Al+3Cu2+ (0.6M) →2Al3+(0.35M)+ Cu (Given:E =2.0V) 0cell

Archive page 3 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 0.75; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 3/7 (5.0 Marks) (5.0 Marks) (6.0 Marks) 2.b. i) What is ion selective electrode? Write the cell representation when a glass electrode is coupled with a saturated calomel electrode. (2M) (ii) When a glass electrode was coupled with a saturated calomel electrode to measure the pH of an unknown solution, an emf of 0.786V was recorded at 298K when pH = 4.2. At the same temperature when a solution of unknown pH was used, an emf of 0.825V was recorded. Calculate the pH of the unknown solution. (Given: E =0.2412V) (3M) 2.c. Two metals such as iron and tin are in contact with each other and exposed to the corrosive environment, name the type of corrosion and explain with the relevant reactions. 2.d. Account for the following: i) Influence of hydrogen over voltage on the rate of corrosion (2M) ii) Anodic corrosion inhibitors (2M) iii) Sacrificial anode method (2M) SCE

EV0000704 · EV0000705 · EV0000706 · +15 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Origin of electrode potential, cell potential transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Origin of electrode potential, cell potential transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Origin of electrode potential, cell potential transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Origin of electrode potential, cell potential transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/001_Origin_of_electrode_potential_cell_potential__3e2916789bf0cc7b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrodes transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrodes transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrodes transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Types of electrodes transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/003_Types_of_electrodes__5665b6defd1e742e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reference electrodes, calomel electrode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Reference electrodes, calomel electrode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Reference electrodes, calomel electrode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.
Reference electrodes, calomel electrode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/004_Reference_electrodes_calomel_electrode__66094e14ed140bda.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Concentration cells, types of concentration cells transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Concentration cells, types of concentration cells transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Concentration cells, types of concentration cells transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Concentration cells, types of concentration cells transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/005_Concentration_cells_types_of_concentration_cells__4982c0baa14e93c6.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electrochemical theory of corrosion transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electrochemical theory of corrosion transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Electrochemical theory of corrosion transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Electrochemical theory of corrosion transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Electrochemical_theory_of_corrosion__b61af161e6bad4a4.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numerical problems based on electrode and cell potential determination transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Numerical problems based on electrode and cell potential determination transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numerical problems based on electrode and cell potential determination transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.
Numerical problems based on electrode and cell potential determination transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__8150689d77526239.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1 transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_differential_aeration_pitting_corrosion_stress_corrosion_-_Part_1__75f77ebe6da9e6f8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1 transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_differential_aeration_pitting_corrosion_stress_corrosion_-_Part_1__75f77ebe6da9e6f8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1 transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_differential_aeration_pitting_corrosion_stress_corrosion_-_Part_1__75f77ebe6da9e6f8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9971; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal, differential aeration, pitting corrosion, stress corrosion - Part 1 transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_differential_aeration_pitting_corrosion_stress_corrosion_-_Part_1__75f77ebe6da9e6f8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9963; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Types of corrosion - differential metal (mention of sacrificial anode corrosion protection), differential aeration, pitting corrosion, stress corrosion - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__eca6ea3854aa0e67.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Types of corrosion - stress corrosion - Part III transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/008_Types_of_corrosion_-_stress_corrosion_-_Part_III__5da0c5dd14cf8164.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.
Factors affecting rate of corrosion (nature of metal, nature of corrosion product, polarization, overvoltage) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__fdef40981a54d818.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_II__bf6644019910a8af.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_II__bf6644019910a8af.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_II__bf6644019910a8af.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_II__bf6644019910a8af.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Corrosion control - metal coating (galvanizing), inorganic coating (anodizing) - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing_-_Part_I__426c13eca70a4fd6.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__54f68b4b385cdc3b.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__54f68b4b385cdc3b.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__54f68b4b385cdc3b.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9963; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part II transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__54f68b4b385cdc3b.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9964; only the JSON segmentation supplies the displayed timestamps.
Corrosion inhibitors, cathodic protection –impressed current method, Use of AI tools in corrosion detection and control - Part I transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/AV_Summaries/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__9d98be19f2c4ea84.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Origin of electrode potential cell potential	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Origin of electrode potential cell potential	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Origin of electrode potential cell potential (converted PDF; original hash 2ec83ffe0ef9ffdc3cf9456eb8d23916168a1766212c76e6ea1d250785e1a9fe)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/001_Origin_of_electrode_potential_cell_potential__c76de2692d228d06.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Nernst equation and its derivation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Nernst equation and its derivation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Nernst equation and its derivation (converted PDF; original hash 901ec34e6c7457fc05d8553aac54951d1d3cf815bd88ea702ba9d98dc08376c9)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/002_Nernst_equation_and_its_derivation__49dac35211b6d743.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Types of electrodes	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Types of electrodes	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Types of electrodes (converted PDF; original hash 48a406f54e08258accb8bb124fb41fa06ec3d46d5d2d0e5677ae46877f62d136)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/003_Types_of_electrodes__ca435e53fd3f8afb.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Reference electrodes calomel electrode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Reference electrodes calomel electrode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Reference electrodes calomel electrode (converted PDF; original hash 523e549f45a848fead9f881858bff0e91e9842ecbf27ad439dd04e5317432d81)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/004_Reference_electrodes_calomel_electrode__25143196395dc884.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Concentration cells types of concentration cells	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Concentration cells types of concentration cells	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Concentration cells types of concentration cells (converted PDF; original hash 0c89dcbfb44000fe7dfaffb7f867bf3434272b52c2c683a1e33e6cea31cd00d9)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/005_Concentration_cells_types_of_concentration_cells__8bdac224f8228816.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Electrochemical theory of corrosion	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Electrochemical theory of corrosion	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Electrochemical theory of corrosion (converted PDF; original hash 06d6b3e6d9112aed3508b8efd82d97e4c6f9335e7bf49c38b421f7d3d89a5332)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Electrochemical_theory_of_corrosion__ba37b2bff7204199.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Numerical problems based on electrode and cell potential determination	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Numerical problems based on electrode and cell potential determination	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Numerical problems based on electrode and cell potential determination (converted PDF; original hash d16e81ca0c88a84a96aa5c6872c372c7a562ff637112bbaa4e3dec1768a2a669)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/006_Numerical_problems_based_on_electrode_and_cell_potential_determination__a4edb9f92dfe8130.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Types of corrosion differential metal mention of sacrificial anode corrosion protection differential aeratio (converted PDF; original hash 3642374cacec4d437aba5c53e58b25c1ef51a91fbcbcfa18224d57dea5962bb3)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/008_Types_of_corrosion_-_differential_metal_mention_of_sacrificial_anode_corrosion_protection_differential_aeratio__e3a8a6acd5ac3c3d.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Factors affecting rate of corrosion nature of metal nature of corrosion product polarization overvoltage (converted PDF; original hash 8a08dc91bae80c3389f9bbd219ecdaa28ea7542d2a73e4d9e864b959e10fd01c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/009_Factors_affecting_rate_of_corrosion_nature_of_metal_nature_of_corrosion_product_polarization_overvoltage__9607117aa91aad2f.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Corrosion control metal coating galvanizing inorganic coating anodizing	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Corrosion control metal coating galvanizing inorganic coating anodizing	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Corrosion control metal coating galvanizing inorganic coating anodizing (converted PDF; original hash f9b7aec8a4e9b902b072e17d3e33cbd18a4b8199c8d22c74c4e48ef58ba6ac57)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/010_Corrosion_control_-_metal_coating_galvanizing_inorganic_coating_anodizing__49b196ca67b756f3.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Corrosion inhibitors cathodic protection impressed current method Use of AI tools in corrosion detection and c (converted PDF; original hash 012f8192c8f8ae555e88e96053fe4b311c1e5b1f31b00ef4aa6f67828573e804)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/011_Corrosion_inhibitors_cathodic_protection_impressed_current_method_Use_of_AI_tools_in_corrosion_detection_and_c__8eeac5bd55474c16.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Experiential learning Based on corrosion chemistry	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Experiential learning Based on corrosion chemistry	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Experiential learning Based on corrosion chemistry (converted PDF; original hash 1f96475bab9ff13a23696c1d32bac7a405682eb92e45eec1a69787638ac47977)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_02/Slides/012_Experiential_learning-_Based_on_corrosion_chemistry__34bf695e05f87f1a.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: 10__1vFS6E1bKV.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/10__1vFS6E1bKV.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 11__1zfeA5C-J7.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/11__1zfeA5C-J7.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 1__1rBrYYqw3s.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/1__1rBrYYqw3s.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 3__1MwY1CTl_j.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/3__1MwY1CTl_j.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 4__1eKMFaJ2lx.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/4__1eKMFaJ2lx.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 5__1pIeTzocnU.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/5__1pIeTzocnU.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 6__1H6QH_PVNf.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/6__1H6QH_PVNf.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 7__16pEjcSTVO.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/7__16pEjcSTVO.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Source	Path	Use
Extracted historical pages: 8__1015pA2nzY.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/8__1015pA2nzY.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 9__14fBQb1-hv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/9__14fBQb1-hv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-EC[2025]__1ZtWpoG9Dv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-EC[2025]__1ZtWpoG9Dv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-RR[2025]__1xN30tqzNj.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-RR[2025]__1xN30tqzNj.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00003	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00005	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00006	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00007	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...
SRC00009	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00010	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00011	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00013	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00014	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00015	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00017	7 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00018	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...
SRC00019	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...
SRC00021	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00022	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00023	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00025	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00026	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00027	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00028	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...
SRC00029	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00034	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00035	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00037	6 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00038	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00039	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00040	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...
SRC00041	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00042	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00043	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00044	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00045	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00050	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00051	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00052	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00053	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00054	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare...
SRC00055	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00056	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare...
SRC00057	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0065. Sidecar variants compare...
SRC00058	0 covered · 0 excluded · 11 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00060	0 covered · 0 excluded · 11 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00059	0 covered · 0 excluded · 11 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00061	6 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00063	6 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00062	6 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00064	0 covered · 0 excluded · 10 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00066	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00065	0 covered · 0 excluded · 10 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00067	0 covered · 0 excluded · 9 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00069	0 covered · 0 excluded · 9 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00068	0 covered · 0 excluded · 9 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00070	0 covered · 0 excluded · 10 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00072	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00071	0 covered · 0 excluded · 10 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00073	0 covered · 0 excluded · 13 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00075	0 covered · 0 excluded · 13 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00074	0 covered · 0 excluded · 13 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00076	0 covered · 0 excluded · 10 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00078	0 covered · 0 excluded · 10 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00077	0 covered · 0 excluded · 10 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00079	0 covered · 0 excluded · 23 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00081	0 covered · 0 excluded · 23 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00080	0 covered · 0 excluded · 23 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00082	0 covered · 0 excluded · 14 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00084	0 covered · 0 excluded · 14 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00083	0 covered · 0 excluded · 14 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00085	0 covered · 0 excluded · 15 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00087	0 covered · 0 excluded · 15 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00086	0 covered · 0 excluded · 15 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00088	0 covered · 0 excluded · 19 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00090	0 covered · 0 excluded · 19 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00089	0 covered · 0 excluded · 19 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00091	6 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00093	6 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00092	6 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00094	3 covered · 0 excluded · 0 gaps	q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed. q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed. q6: Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
SRC00095	1 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
SRC00096	1 covered · 0 excluded · 0 gaps	q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
SRC00097	1 covered · 0 excluded · 0 gaps	q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
SRC00098	2 covered · 0 excluded · 0 gaps	q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed. q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
SRC00099	1 covered · 0 excluded · 0 gaps	q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.

Source	Unit counts	Example decisions
SRC00100	2 covered · 0 excluded · 0 gaps	q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed. q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
SRC00101	1 covered · 0 excluded · 0 gaps	q4: Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
SRC00102	1 covered · 0 excluded · 0 gaps	q5: Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
SRC00103	1 covered · 0 excluded · 0 gaps	q6: Original archive PDF page 6; candidate unit association is text-based and unconfirmed.
SRC00104	2 covered · 0 excluded · 0 gaps	q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed. q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
SRC00105	2 covered · 0 excluded · 0 gaps	q2: Original archive PDF page 2; candidate unit association is text-based and unconfirmed. q3: Original archive PDF page 3; candidate unit association is text-based and unconfirmed.

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